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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Magnetic Effect of Current

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 211 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{\hat{k}+\hat{i}}{\sqrt{2}}$ is placed in a magnetic field, $\vec{B} = B_o(3\hat{i} + 2\hat{k})$. If $B_o = 4 \times 10^{-3}$ T and current $I = 100\sqrt{2}$ A, the torque experienced by the loop is ____ Wb.A. ($\pi = 3.14$)

(A) $16 \times 10^{-5}\hat{k}$ (B) $5024 \times 10^{-7}\hat{k}$ (C) $5024 \times 10^{-7}\hat{i}$ (D) $5024 \times 10^{-7}\hat{j}$

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

A current of 30 A each flows in opposite directions in two conducting wires, placed parallel to each other at a distance of 8 cm . The magnetic field at the mid point between the two wires is ____ μT. $\left(\frac{\mu_o}{4\pi} = 10^{-7}\ \text{N/A}^2\right)$

(A) 30 (B) 300 (C) 150 (D) 0.0

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

A small cube of side 1 mm is placed at the centre of a circular loop of radius 10 cm carrying a current of 2 A . The magnetic energy stored inside the cube is $\alpha \times 10^{-14}$ J. The value of α is ____ . $\left(\mu_o = 4\pi \times 10^{-7}\text{Tm/A}, \pi = 3.14\right)$

(A) 6.28 (B) 6.28×10^{-6} (C) 628 (D) 6.28×10^{-4}

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

A particle of charge q and mass m is projected from origin with an initial velocity $\vec{v} = \left(\frac{v_0}{\sqrt{2}}\hat{x} + \frac{v_0}{\sqrt{2}}\hat{y}\right)$. There exists a uniform magnetic field $\vec{B} = B_0\hat{z}$ and a space varying electric field $\vec{E} = E_o e^{-\lambda x}\hat{x}$ within the region $0 \leqslant x \leqslant L$. After travelling a distance such that x-coordinate has changed from $x = 0$ to $x = L$, the change in the kinetic energy is ____ .

(A) $\frac{qE_0}{\lambda}\left[1 - e^{-\lambda L}\right]$ (B) $\left(\frac{v_0 qB_0}{2\lambda}\right)\left[2 - e^{-2\lambda L}\right]$

(C) $\frac{qE_0}{\lambda}\left[1 + e^{-\lambda L}\right]$ (D) $q\left(\frac{E_0 + v_0 B_0}{\lambda}\right)\left[1 - e^{-\lambda L/2}\right]$

Open solution

Q5. JEE Main 2026 (Online) 4th April Morning Shift

An insulated wire is wound so that it forms a flat coil with $N = 200$ turns. The radius of the innermost turn is $r_1 = 3$ cm, and of the outermost turn $r_2 = 6$ cm. If 20 mA current flows in it then the magnetic moment will be $\alpha \times 10^{-2}$ A. m^2. The value of α is ____ .

(A) 4.4 (B) 2.64 (C) 3.25 (D) 1.2

Open solution

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Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

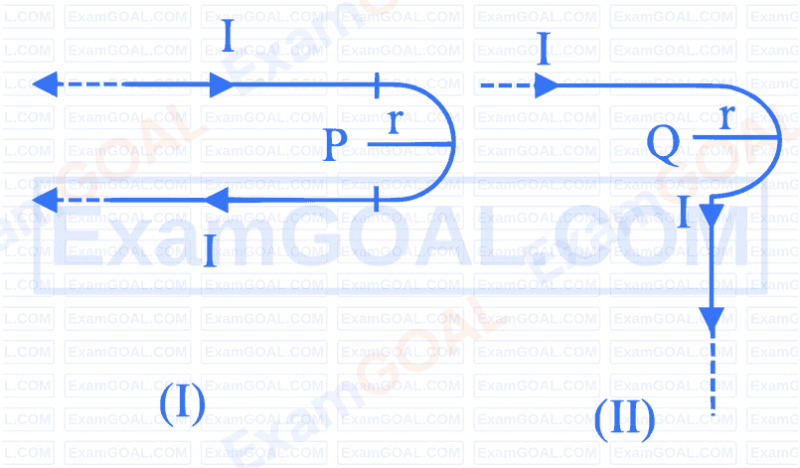
A particle having charge 10^{-9} C moving in x-y plane in fields of $0.4\hat{j}$ N/C and $4 \times 10^{-3}\hat{k}$ T experiences a force of $(4\hat{i} + 2\hat{j}) \times 10^{-10}$ N. The velocity of the particle at that instant is _____ m/s.

(A) $50\hat{i} + 100\hat{j}$ (B) $100\hat{i} + 50\hat{j}$ (C) $-50\hat{i} + 100\hat{j}$ (D) $50\hat{i} - 100\hat{j}$

Q7. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Two identical long current carrying wires are bent into the shapes shown in the following figures. If the magnitude of magnetic fields at the centres P and Q of a semicircular arc are B_1 and B_2 respectively, then the ratio $\frac{B_1}{B_2}$ is _____.



(A) $\frac{2+\pi}{1+\pi}$ (B) $\frac{1+\pi}{1-\pi}$ (C) $\frac{2+\pi}{1-\pi}$ (D) $\frac{1+\pi}{2-\pi}$

Q8. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

A long cylindrical conductor with large cross section carries an electric current distributed uniformly over its cross-section. Magnetic field due to this current is:

A. maximum at either ends of the conductor and minimum at the midpoint

B. maximum at the axis of the conductor

C. minimum at the surface of the conductor

D. minimum at the axis of the conductor

E. same at all points in the cross-section of the conductor

Choose the **correct** answer from the options given below:

(A) B, C Only (B) E Only (C) A, D Only (D) D Only

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Q9. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

The magnetic field at the centre of a current carrying circular loop of radius R is 16μ T. The magnetic field at a distance $x = \sqrt{3}R$ on its axis from the centre is

____ μT.

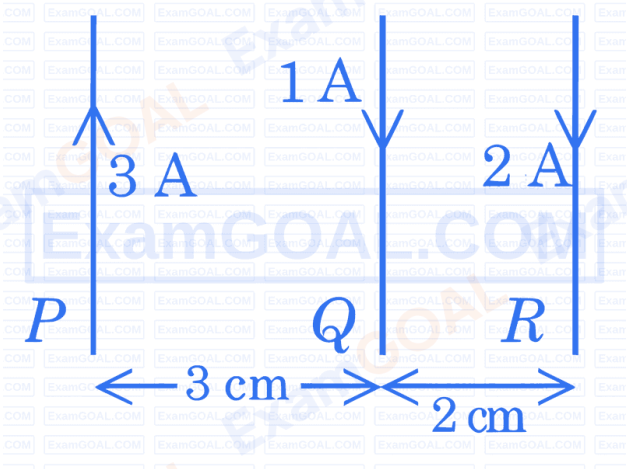
(A) 8 (B) 2 (C) 4 (D) $2\sqrt{2}$

Q10. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Three long straight wires carrying current are arranged mutually parallel as shown in the figure. The force experienced by 15 cm length of wire Q is ____ .

$(\mu_o = 4\pi \times 10^{-7}\ \mathrm{T.m/A})$



(A) 6×10^{-6} N towards P (B) 6×10^{-6} N towards R

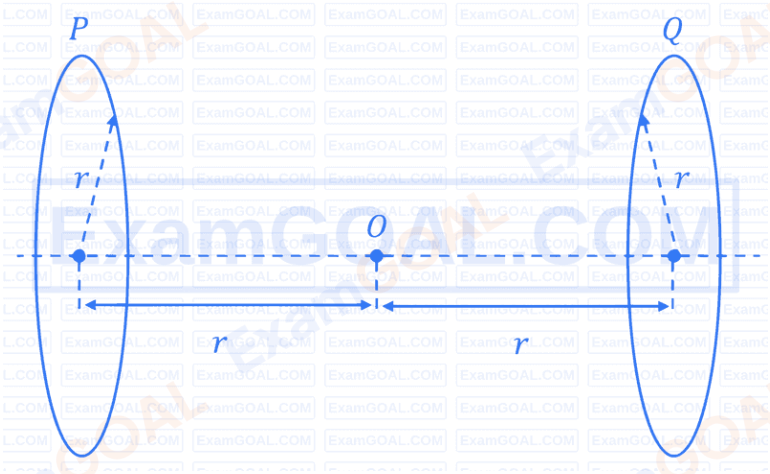
(C) 6×10^{-7} N towards P (D) 6×10^{-7} N towards R

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Q11. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O. The net magnetic field at O is :



(A) $\frac{\mu_o I}{4\sqrt{2}r}$ towards Q

(B) $\frac{3\mu_o I}{4\sqrt{2}r}$ towards Q

(C) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards P

(D) $\frac{\mu_o I}{4\sqrt{2}r}$ towards P

Q12. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

The current passing through a conducting loop in the form of equilateral triangle of side $4\sqrt{3}$ cm is 2 A . The magnetic field at its centroid is $\alpha \times 10^{-5}$ T. The value of α is ____ .

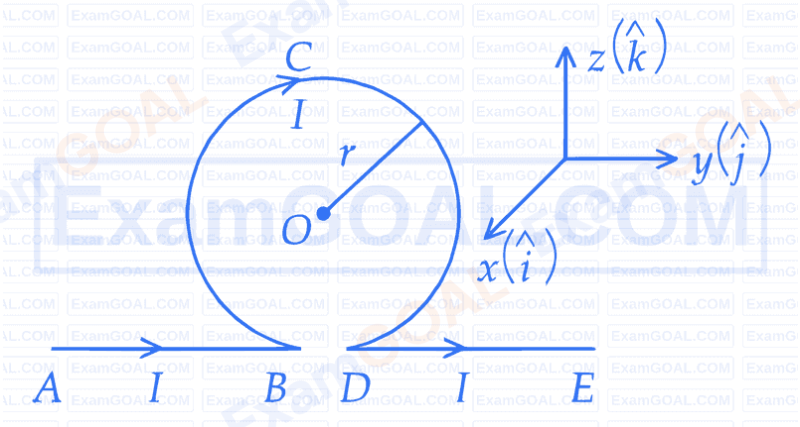
(Given : $\mu_o = 4\pi \times 10^{-7}$ SI units)

(A) $3\sqrt{3}$ (B) $2\sqrt{3}$ (C) $\sqrt{3}$ (D) $\frac{\sqrt{3}}{2}$

Q13. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

An infinitely long straight wire carrying current I is bent in a planar shape as shown in the diagram. The radius of the circular part is r. The magnetic field at the centre O of the circular loop is :



(A) $-\frac{\mu_0}{2\pi}\frac{I}{r}(\pi+1)\hat{i}$

(B) $-\frac{\mu_0}{2\pi}\frac{I}{r}(\pi-1)\hat{i}$

(C) $\frac{\mu_0}{2\pi}\frac{I}{r}(\pi+1)\hat{i}$

(D) $\frac{\mu_0}{2\pi}\frac{I}{r}(\pi-1)\hat{i}$

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Q14. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

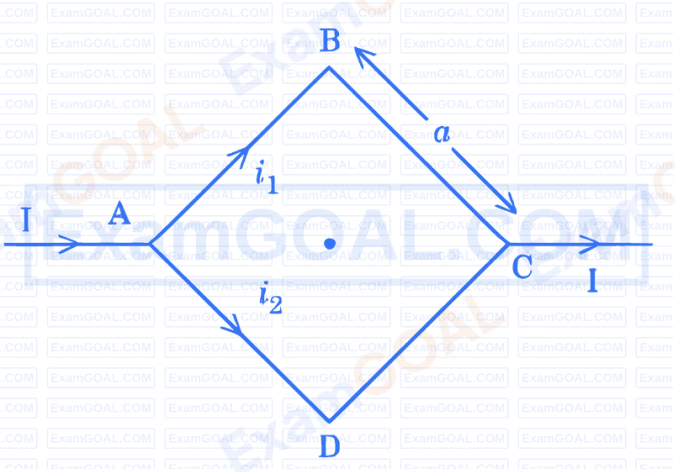
A current carrying solenoid is placed vertically and a particle of mass m with charge Q is released from rest. The particle moves along the axis of solenoid. If g is acceleration due to gravity then the acceleration (a) of the charged particle will satisfy :

(A) $a > g$ (B) $a = 0$ (C) $0 < a < g$ (D) $a = g$

Q15. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Figure shows a current carrying square loop ABCD of edge length 'a' lying in a plane. If the resistance of the ABC part is r and that of ADC part is 2r, then the magnitude of the resultant magnetic field at centre of the square loop is



(A) $\frac{2\mu_0 I}{3\pi a}$ (B) $\frac{\sqrt{2}\mu_0 I}{3\pi a}$ (C) $\frac{3\pi\mu_0 I}{\sqrt{2}a}$ (D) $\frac{\mu_0 I}{2\pi a}$

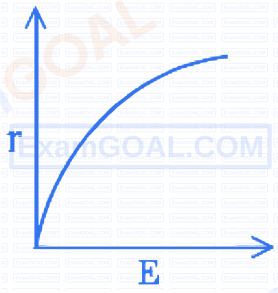
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Q16. JEE Main 2025 (Online) 7th April Morning Shift

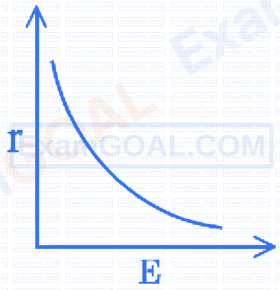
Open solution

A particle of charge q, mass m and kinetic energy E enters in magnetic field perpendicular to its velocity and undergoes a circular arc of radius (r). Which of the following curves represents the variation of r with E ?

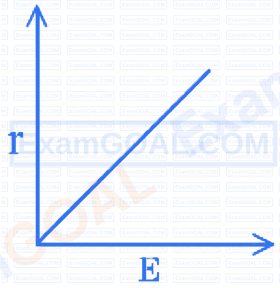
(A)



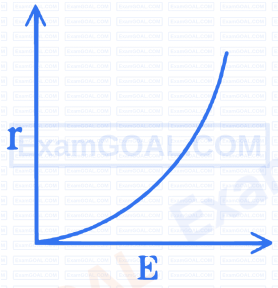
(B)



(C)



(D)



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Q17. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Uniform magnetic fields of different strengths (B_1 and B_2), both normal to the plane of the paper exist as shown in the figure. A charged particle of mass m and charge q, at the interface at an instant, moves into the region 2 with velocity v and returns to the interface. It continues to move into region 1 and finally reaches the interface. What is the displacement of the particle during this movement along the interface?



(Consider the velocity of the particle to be normal to the magnetic field and $B_2 > B_1$)

(A) $\frac{mv}{qB_1}\left(1-\frac{B_1}{B_2}\right)\times 2$

(B) $\frac{mv}{qB_1}\left(1-\frac{B_2}{B_1}\right)\times 2$

(C) $\frac{mv}{qB_1}\left(1-\frac{B_2}{B_1}\right)$

(D) $\frac{mv}{qB_1}\left(1-\frac{B_1}{B_2}\right)$

Q18. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A : If Oxygen ion $\left(O^{-2}\right)$ and Hydrogen ion (H^+)enter normal to the magnetic field with equal momentum, then the path of O^{-2} ion has a smaller curvature than that of H^+.

Reason R : A proton with same linear momentum as an electron will form a path of smaller radius of curvature on entering a uniform magnetic field perpendicularly.

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

Q19. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

In a moving coil galvanometer, two moving coils M_1 and M_2 have the following particulars :

$R_1 = 5\Omega,\ N_1 = 15,\ A_1 = 3.6\times 10^{-3}\ m^2,\ B_1 = 0.25\ T$

$R_2 = 7\Omega,\ N_2 = 21,\ A_2 = 1.8\times 10^{-3}\ m^2,\ B_2 = 0.50\ T$

Assuming that torsional constant of the springs are same for both coils, what will be the ratio of voltage sensitivity of M_1 and M_2 ?

(A) $1:1$ (B) $1:3$ (C) $1:2$ (D) $1:4$

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Q20. JEE Main 2025 (Online) 2nd April Morning Shift

Let B_1 be the magnitude of magnetic field at center of a circular coil of radius R carrying current I. Let B_2 be the magnitude of magnetic field at an axial distance ' x ' from the center. For $x : R = 3 : 4$, $\frac{B_2}{B_1}$ is :

Open solution

(A) $64 : 125$ (B) $25 : 16$ (C) $4 : 5$ (D) $16 : 25$

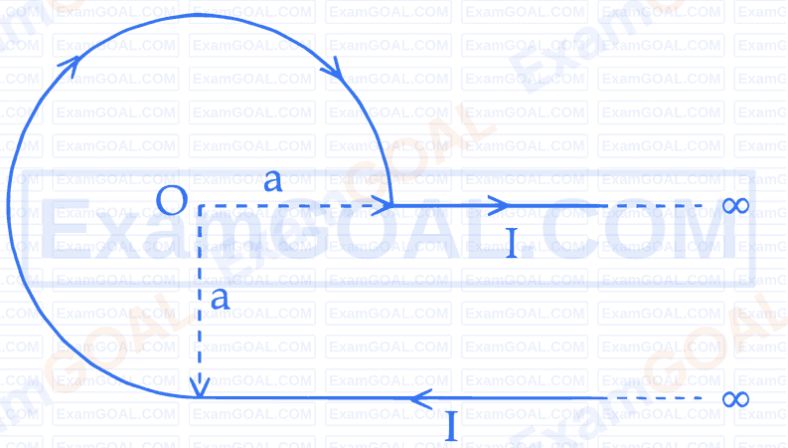
Q21. JEE Main 2025 (Online) 29th January Morning Shift

Consider a long straight wire of a circular cross-section (radius *a*) carrying a steady current *I*. The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the wire, outside the wire] is half of the maximum possible magnetic field, any where due to the wire, will be :

Open solution

(A) [*a*/2, 3*a*] (B) [*a*/4, 3*a*/2] (C) [*a*/2, 2*a*] (D) [*a*/4, 2*a*]

Q22. JEE Main 2025 (Online) 28th January Evening Shift



Open solution

An infinite wire has a circular bend of radius *a*, and carrying a current *I* as shown in the figure. The magnitude of magnetic field at the origin O of the arc is given by:

(A) $\frac{\mu_0}{2\pi}\frac{I}{a}\left[\frac{\pi}{2}+2\right]$ (B) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{3\pi}{2}+1\right]$

(C) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{\pi}{2}+1\right]$ (D) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{3\pi}{2}+2\right]$

Q23. JEE Main 2025 (Online) 28th January Morning Shift

Consider a long thin conducting wire carrying a uniform current I. A particle having mass " m " and charge " q " is released at a distance " a " from the wire with a speed v_0 along the direction of current in the wire. The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is [μ_0 is vacuum permeability]

Open solution

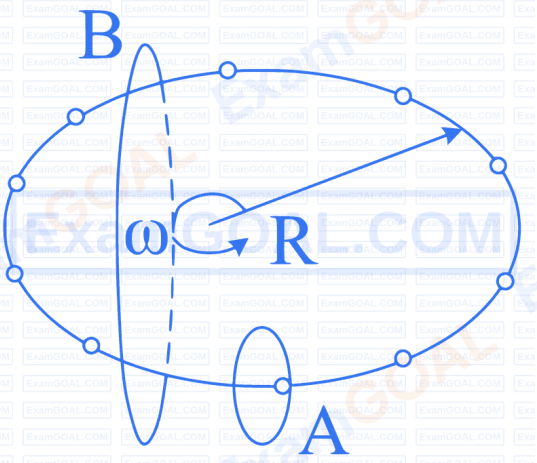
(A) $a\left[1-\frac{mv_o}{2q\mu_0 I}\right]$ (B) $ae^{-\frac{4\pi mv_o}{q\mu_o I}}$

(C) $a\left[1-\frac{mv}{q}\mu_o I\right]$ (D) $\frac{a}{2}$

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Q24. JEE Main 2025 (Online) 24th January Evening Shift

Open solution



N equally spaced charges each of value q , are placed on a circle of radius R . The circle rotates about its axis with an angular velocity ω as shown in the figure. A bigger Amperian loop B encloses the whole circle where as a smaller Amperian loop A encloses a small segment. The difference between enclosed currents, $I_A - I_B$, for the given Amperian loops is

(A) $\frac{N^2}{2\pi}q\omega$

(B) $\frac{N}{2\pi}q\omega$

(C) $\frac{2\pi}{N}q\omega$

(D) $\frac{N}{\pi}q\omega$

Q25. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : An electron in a certain region of uniform magnetic field is moving with constant velocity in a straight line path.

Reason (R) : The magnetic field in that region is along the direction of velocity of the electron. In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true but $(\mathbf{R})$ is NOT the correct explanation of (A)

(D) (A) is false but (R) is true

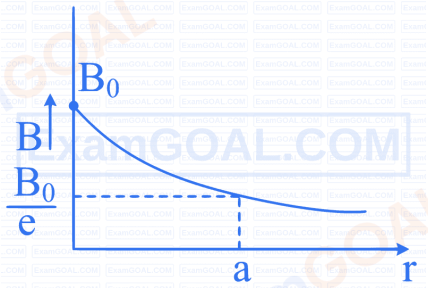
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Q26. JEE Main 2025 (Online) 24th January Evening Shift

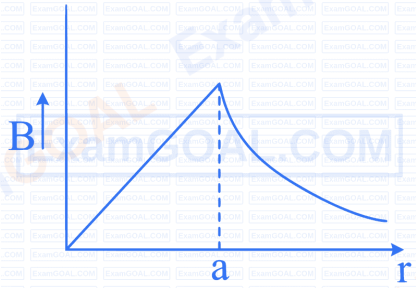
Open solution

A long straight wire of a circular cross-section with radius ' a ' carries a steady current I . The current I is uniformly distributed across this cross-section. The plot of magnitude of magnetic field B with distance r from the centre of the wire is given by

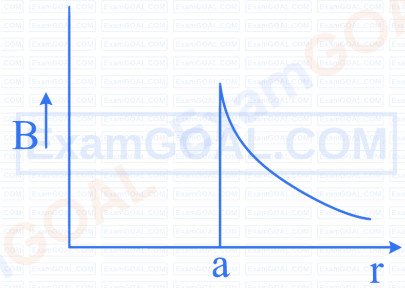
(A)



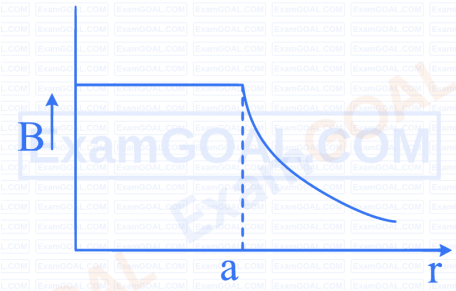
(B)



(C)



(D)



Q27. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

An electron projected perpendicular to a uniform magnetic field B moves in a circle. If Bohr's quantization is applicable, then the radius of the electronic orbit in the first excited state is :

(A) $\sqrt{\frac{h}{\pi eB}}$

(B) $\sqrt{\frac{4h}{\pi eB}}$

(C) $\sqrt{\frac{\mathrm{h}}{2\pi \mathrm{eB}}}$

(D) $\sqrt{\frac{2h}{\pi eB}}$

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Q28. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

A proton and a deuteron $(q = +e, m = 2.0u)$ having same kinetic energies enter a region of uniform magnetic field $\vec{B}$, moving perpendicular to $\vec{B}$. The ratio of the radius r_d of deuteron path to the radius r_p of the proton path is:

(A) $1:2$ (B) $1:1$ (C) $\sqrt{2}:1$ (D) $1:\sqrt{2}$

Q29. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Given below are two statements :

Statement (I) : When currents vary with time, Newton's third law is valid only if momentum carried by the electromagnetic field is taken into account

Statement (II) : Ampere's circuital law does not depend on Biot-Savart's law.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q30. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

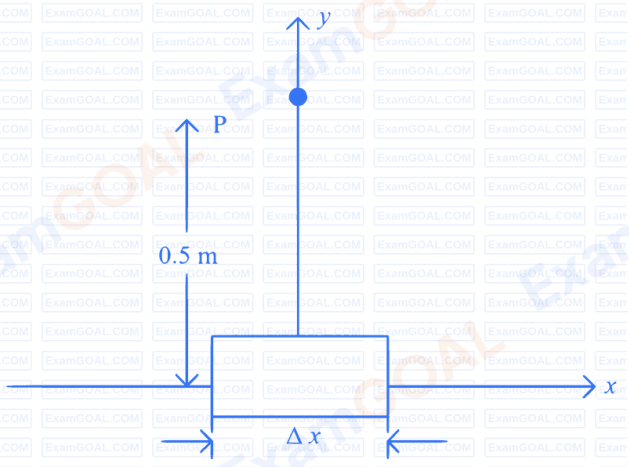
A long straight wire of radius a carries a steady current I. The current is uniformly distributed across its cross section. The ratio of the magnetic field at $\frac{a}{2}$ and $2a$ from axis of the wire is :

(A) $4:1$ (B) $3:4$ (C) $1:1$ (D) $1:4$

Q31. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

An element $\Delta l = \Delta x\hat{i}$ is placed at the origin and carries a large current $I = 10\ \text{A}$. The magnetic field on the y-axis at a distance of $0.5\ \text{m}$ from the elements Δx of $1\ \text{cm}$ length is:



(A) $10 \times 10^{-8}\ \text{T}$ (B) $8 \times 10^{-8}\ \text{T}$ (C) $4 \times 10^{-8}\ \text{T}$ (D) $12 \times 10^{-8}\ \text{T}$

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Q32. JEE Main 2024 (Online) 5th April Evening Shift

The electrostatic force $\left(\vec{F}_1\right)$ and magnetic force $\left(\vec{F}_2\right)$ acting on a charge q moving with velocity v can be written :

Open solution

(A) $\vec{F}_1 = q\vec{B}, \vec{F}_2 = q(\vec{B} \times \vec{v})$

(B) $\vec{F}_1 = q\vec{V} \cdot \vec{E}, \vec{F}_2 = q(\vec{B} \cdot \vec{V})$

(C) $\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{V} \times \vec{B})$

(D) $\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{B} \times \vec{V})$

Q33. JEE Main 2024 (Online) 5th April Morning Shift

In a co-axial straight cable, the central conductor and the outer conductor carry equal currents in opposite directions. The magnetic field is zero :

Open solution

(A) inside the outer conductor

(B) outside the cable

(C) in between the two conductors

(D) inside the inner conductor

Q34. JEE Main 2024 (Online) 4th April Morning Shift

An electron is projected with uniform velocity along the axis inside a current carrying long solenoid. Then :

Open solution

(A) the electron will experience a force at 45° to the axis and execute a helical path.

(B) the electron will be accelerated along the axis.

(C) the electron path will be circular about the axis.

(D) the electron will continue to move with uniform velocity along the axis of the solenoid.

Q35. JEE Main 2024 (Online) 31st January Evening Shift

A uniform magnetic field of $2 \times 10^{-3}\ \mathrm{T}$ acts along positive Y-direction. A rectangular loop of sides $20\ \mathrm{cm}$ and $10\ \mathrm{cm}$ with current of $5\ \mathrm{A}$ is in $Y - Z$ plane. The current is in anticlockwise sense with reference to negative X axis. Magnitude and direction of the torque is:

Open solution

(A) $2 \times 10^{-4}\ \mathrm{N\text{-}m}$ along negative Z-direction

(B) $2 \times 10^{-4}\ \mathrm{N\text{-}m}$ along positive X-direction

(C) $2 \times 10^{-4}\ \mathrm{N\text{-}m}$ along positive Y-direction

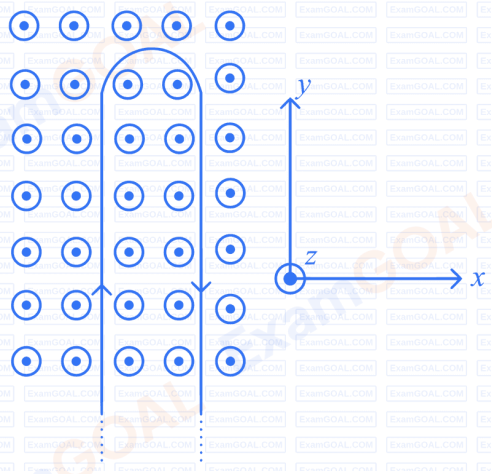
(D) $2 \times 10^{-4}\ \mathrm{N\text{-}m}$ along positive Z-direction

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Q36. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

A rigid wire consists of a semicircular portion of radius R and two straight sections. The wire is partially immersed in a perpendicular magnetic field $B = B_0\hat{k}$ as shown in figure. The magnetic force on the wire if it has a current i is:

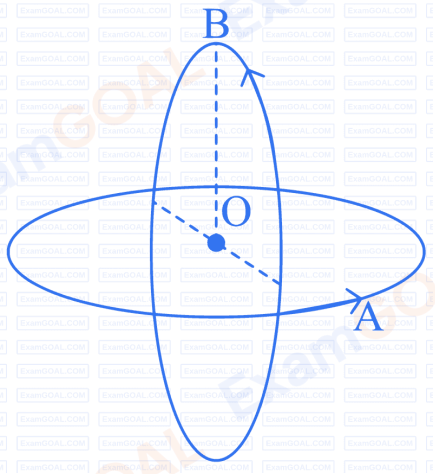


(A) $iBR\hat{j}$ (B) $-2iBR\hat{j}$ (C) $2iBR\hat{j}$ (D) $-iBR\hat{j}$

Q37. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Two insulated circular loops A and B of radius 'a' carrying a current of 'I' in the anti clockwise direction as shown in the figure. The magnitude of the magnetic induction at the centre will be :



(A) $\frac{\sqrt{2}\mu_0 I}{a}$ (B) $\frac{\mu_0 I}{\sqrt{2}a}$
(C) $\frac{\mu_0 \mathrm{I}}{2\mathrm{a}}$ (D) $\frac{2\mu_0 I}{a}$

Q38. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

Two particles X and Y having equal charges are being accelerated through the same potential difference. Thereafter they enter normally in a region of uniform magnetic field and describe circular paths of radii R_1 and R_2 respectively. The mass ratio of X and Y is :

(A) $\left(\frac{R_1}{R_2}\right)$ (B) $\left(\frac{R_2}{R_1}\right)$ (C) $\left(\frac{R_2}{R_1}\right)^2$ (D) $\left(\frac{R_1}{R_2}\right)^2$

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Q39. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

A proton moving with a constant velocity passes through a region of space without any change in its velocity. If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic fields respectively, then the region of space may have :

(A) $E = 0,\ B = 0$

(B) $E = 0,\ B \neq 0$

(C) $E \neq 0,\ B = 0$

(D) $E \neq 0,\ B \neq 0$

Choose the most appropriate answer from the options given below :

(A) (A), (B) and (C) only

(B) (A), (C) and (D) only

(C) (A), (B) and (D) only

(D) (B), (C) and (D) only

Q40. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

An electron is moving along the positive x-axis. If the uniform magnetic field is applied parallel to the negative z-axis, then

A. The electron will experience magnetic force along positive y-axis

B. The electron will experience magnetic force along negative y-axis

C. The electron will not experience any force in magnetic field

D. The electron will continue to move along the positive x-axis

E. The electron will move along circular path in magnetic field

Choose the correct answer from the options given below:

(A) A and E only

(B) B and D only

(C) B and E only

(D) C and D only

Q41. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

The source of time varying magnetic field may be

(A) a permanent magnet

(B) an electric field changing linearly with time

(C) direct current

(D) a decelerating charge particle

(E) an antenna fed with a digital signal

Choose the correct answer from the options given below:

(A) (D) only

(B) (A) only

(C) (B) and (D) only

(D) (C) and (E) only

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Q42. JEE Main 2023 (Online) 11th April Evening Shift

An electron is allowed to move with constant velocity along the axis of current carrying straight solenoid.

A. The electron will experience magnetic force along the axis of the solenoid.

B. The electron will not experience magnetic force.

C. The electron will continue to move along the axis of the solenoid.

D. The electron will be accelerated along the axis of the solenoid.

E. The electron will follow parabolic path-inside the solenoid.

Choose the correct answer from the options given below:

(A) B, C and D only (B) B and C only (C) A and D only (D) B and E only

Open solution

Q43. JEE Main 2023 (Online) 8th April Morning Shift

A charge particle moving in magnetic field B, has the components of velocity along B as well as perpendicular to B. The path of the charge particle will be

(A) helical path with the axis along magnetic field B

(B) straight along the direction of magnetic field B

(C) circular path

(D) helical path with the axis perpendicular to the direction of magnetic field B

Open solution

Q44. JEE Main 2023 (Online) 6th April Morning Shift

A long straight wire of circular cross-section (radius a) is carrying steady current I. The current I is uniformly distributed across this cross-section. The magnetic field is

(A) uniform in the region $r < a$ and inversely proportional to distance r from the axis, in the region $r > a$

(B) zero in the region $r < a$ and inversely proportional to r in the region $r > a$

(C) directly proportional to r in the region $r < a$ and inversely proportional to r in the region $r > a$

(D) inversely proportional to r in the region $r < a$ and uniform throughout in the region $r > a$

Open solution

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Q45. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

As shown in the figure, a long straight conductor with semicircular arc of radius $\frac{\pi}{10}$m is carrying current $\mathrm{I} = 3\mathrm{A}$. The magnitude of the magnetic field, at the center O of the arc is :

(The permeability of the vacuum $= 4\pi \times 10^{-7}\ \mathrm{NA}^{-2}$)

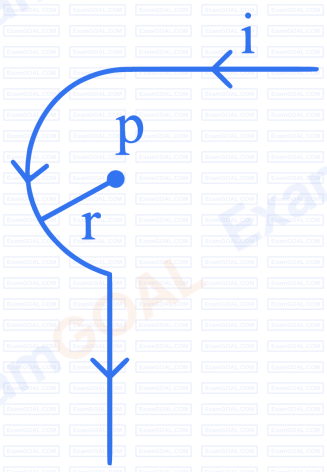


(A) $4\mu\mathrm{T}$ (B) $3\mu\mathrm{T}$ (C) $6\mu\mathrm{T}$ (D) $1\mu\mathrm{T}$

Q46. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

Find the magnetic field at the point P in figure. The curved portion is a semicircle connected to two long straight wires.



(A) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$ (B) $\frac{\mu_0 i}{2r}\left(1+\frac{2}{\pi}\right)$

(C) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{\pi}\right)$ (D) $\frac{\mu_0 i}{2r}\left(1+\frac{1}{\pi}\right)$

Q47. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

A long conducting wire having a current I flowing through it, is bent into a circular coil of N turns. Then it is bent into a circular coil of n turns. The magnetic field is calculated at the centre of coils in both the cases. The ratio of the magnetic field in first case to that of second case is :

(A) $N^2 : n^2$ (B) $\mathrm{N} : \mathrm{n}$ (C) $\mathrm{n} : \mathrm{N}$ (D) $n^2 : N^2$

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Q48. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

A rod with circular cross-section area $2\ \text{cm}^2$ and length $40\ \text{cm}$ is wound uniformly with 400 turns of an insulated wire. If a current of $0.4\ \text{A}$ flows in the wire windings, the total magnetic flux produced inside windings is $4\pi \times 10^{-6}\ \text{Wb}$. The relative permeability of the rod is

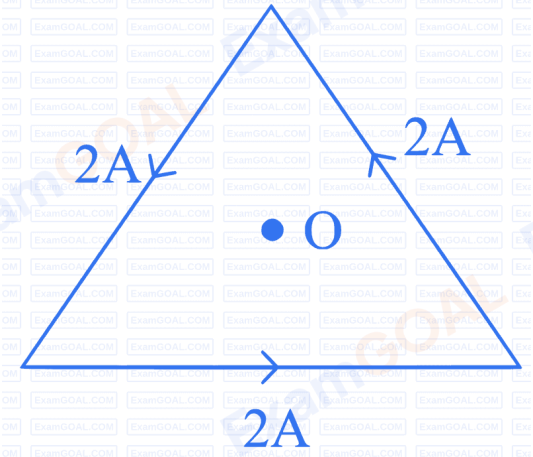
(Given : Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}\text{NA}^{-2}$)

(A) $\frac{5}{16}$ (B) 125 (C) $\frac{32}{5}$ (D) 12.5

Q49. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

As shown in the figure, a current of $2\ \text{A}$ flowing in an equilateral triangle of side $4\sqrt{3}\ \text{cm}$. The magnetic field at the centroid O of the triangle is



(Neglect the effect of earth's magnetic field)

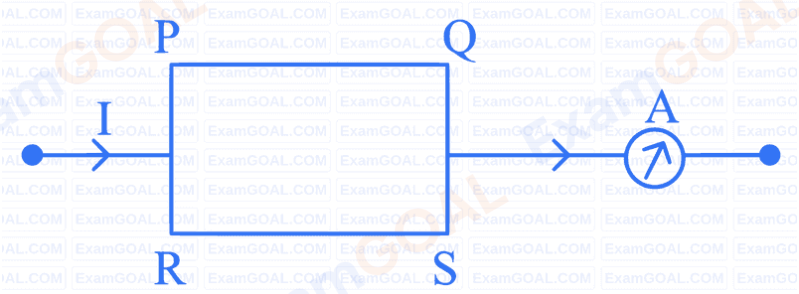
(A) $4\sqrt{3} \times 10^{-4}\ \text{T}$ (B) $4\sqrt{3} \times 10^{-5}\ \text{T}$

(C) $3\sqrt{3} \times 10^{-5}\ \text{T}$ (D) $\sqrt{3} \times 10^{-4}\ \text{T}$

Q50. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

A current carrying rectangular loop PQRS is made of uniform wire. The length $PR = QS = 5\ \text{cm}$ and $PQ = RS = 100\ \text{cm}$. If ammeter current reading changes from I to $2I$, the ratio of magnetic forces per unit length on the wire PQ due to wire RS in the two cases respectively $\left(f_{PQ}^{I} : f_{PQ}^{2I}\right)$ is:



(A) 1 : 4 (B) 1 : 3 (C) 1 : 2 (D) 1 : 5

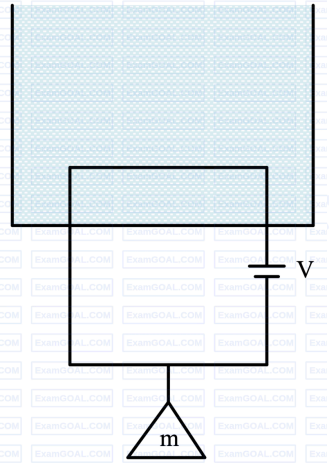
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Q51. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

A massless square loop, of wire of resistance 10Ω, supporting a mass of $1\ \mathrm{g}$, hangs vertically with one of its sides in a uniform magnetic field of $10^3\mathrm{G}$, directed outwards in the shaded region. A dc voltage V is applied to the loop. For what value of V, the magnetic force will exactly balance the weight of the supporting mass of $1\ \mathrm{g}$?

(If sides of the loop $= 10\ \mathrm{cm},\ \mathrm{g} = 10\ \mathrm{ms}^{-2}$)



(A) 1 V (B) $\frac{1}{10}$ V (C) 10 V (D) 100 V

Q52. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

The magnetic moments associated with two closely wound circular coils A and B of radius $\mathrm{r_A} = 10\ \mathrm{cm}$ and $\mathrm{r_B} = 20\ \mathrm{cm}$ respectively are equal if : (Where $\mathrm{N_A}, \mathrm{I_A}$ and $\mathrm{N_B}, \mathrm{I_B}$ are number of turns and current of A and B respectively)

(A) $4\,\mathrm{N_A I_A} = \mathrm{N_B I_B}$

(B) $2\,\mathrm{N_A I_A} = \mathrm{N_B I_B}$

(C) $\mathrm{N_A} = 2\,\mathrm{N_B}$

(D) $\mathrm{N_A I_A} = 4\,\mathrm{N_B I_B}$

Q53. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

The electric current in a circular coil of four turns produces a magnetic induction 32 T at its centre. The coil is unwound and is rewound into a circular coil of single turn, the magnetic induction at the centre of the coil by the same current will be :

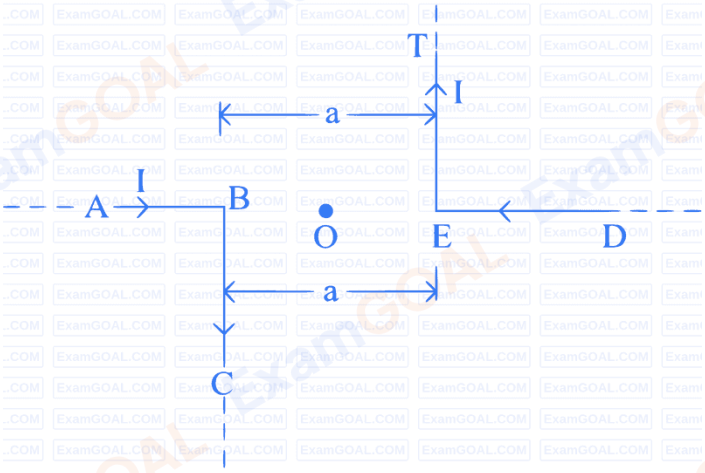
(A) 2 T (B) 4 T (C) 8 T (D) 16 T

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Q54. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

The magnitude of magnetic induction at mid point O due to current arrangement as shown in Fig will be



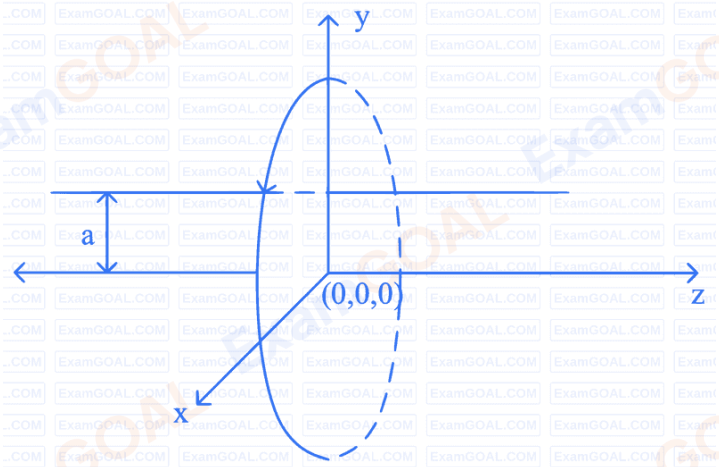
(A) $\frac{\mu_0 I}{\pi a}$ (B) $\frac{\mu_0 I}{4\pi a}$ (C) $\frac{\mu_0 I}{2\pi a}$ (D) 0

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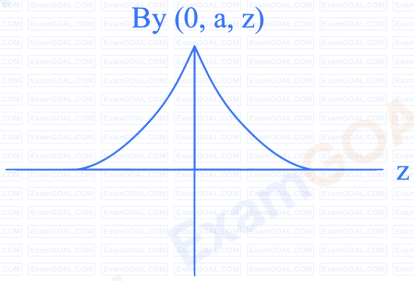
Q55. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

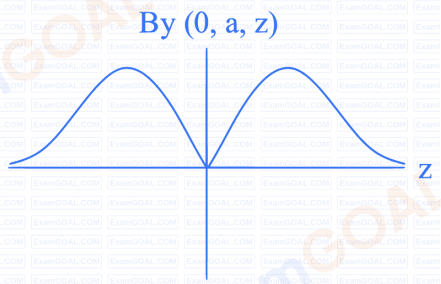
A single current carrying loop of wire carrying current I flowing in anticlockwise direction seen from +ve z direction and lying in xy plane is shown in figure. The plot of $\hat{j}$ component of magnetic field (By) at a distance 'a' (less than radius of the coil) and on yz plane vs z coordinate looks like



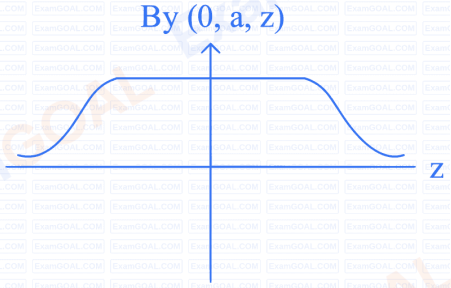
(A)



(B)

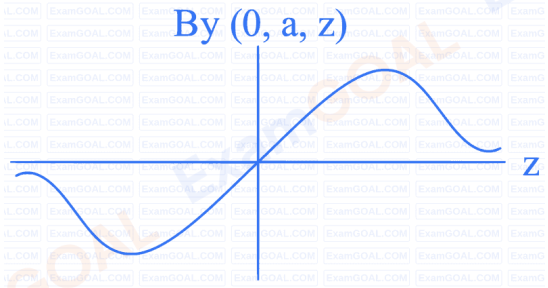


(C)



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(D)



Q56. JEE Main 2023 (Online) 25th January Evening Shift

For a moving coil galvanometer, the deflection in the coil is 0.05 rad when a current of 10 mA is passes through it. If the torsional constant of suspension wire is $4.0 \times 10^{-5} \mathrm{N\,m\,rad^{-1}}$, the magnetic field is 0.01T and the number of turns in the coil is 200, the area of each turn (in $\mathrm{cm^2}$) is :

Open solution

(A) 1.5 (B) 2.0 (C) 0.5 (D) 1.0

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Q57. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Match List I with List II

	List I (Current configuration)		List II (Magnitude of Magnetic Field at point O)
A.		I.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 2]$
B.		II.	$B_0 = \frac{\mu_0}{4}\frac{I}{r}$
C.		III.	$B_0 = \frac{\mu_0 I}{2\pi r}[\pi - 1]$
D.		IV.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 1]$

Choose the correct answer from the options given below :

(A) A-III, B-IV, C-I, D-II

(B) A-II, B-I, C-IV, D-III

(C) A-III, B-I, C-IV, D-II

(D) A-I, B-III, C-IV, D-II

Q58. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

A long solenoid is formed by winding 70 turns cm^{-1}. If 2.0 A current flows, then the magnetic field produced inside the solenoid is _____ ($\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1})

(A) 88×10^{-4} T

(B) 1232×10^{-4} T

(C) 176×10^{-4} T

(D) 352×10^{-4} T

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Q59. JEE Main 2023 (Online) 24th January Morning Shift

Two long straight wires P and Q carrying equal current 10A each were kept parallel to each other at 5 cm distance. Magnitude of magnetic force experienced by 10 cm length of wire P is F_1. If distance between wires is halved and currents on them are doubled, force F_2 on 10 cm length of wire P will be:

Open solution

(A) $\frac{F_1}{8}$ (B) 10 F_1 (C) $\frac{F_1}{10}$ (D) 8 F_1

Q60. JEE Main 2023 (Online) 24th January Morning Shift

A circular loop of radius r is carrying current I A. The ratio of magnetic field at the center of circular loop and at a distance r from the center of the loop on its axis is :

Open solution

(A) $3\sqrt{2}:2$ (B) $1:3\sqrt{2}$ (C) $2\sqrt{2}:1$ (D) $1:\sqrt{2}$

Q61. JEE Main 2022 (Online) 30th July Evening Shift

A wire of length $8\ \mathrm{m}$ is bent to form a circular loop in y-z plane. A current of $0.5\ \mathrm{A}$ is flowing in anticlockwise direction in it, the magnetic dipole moment of the loop is:

Open solution

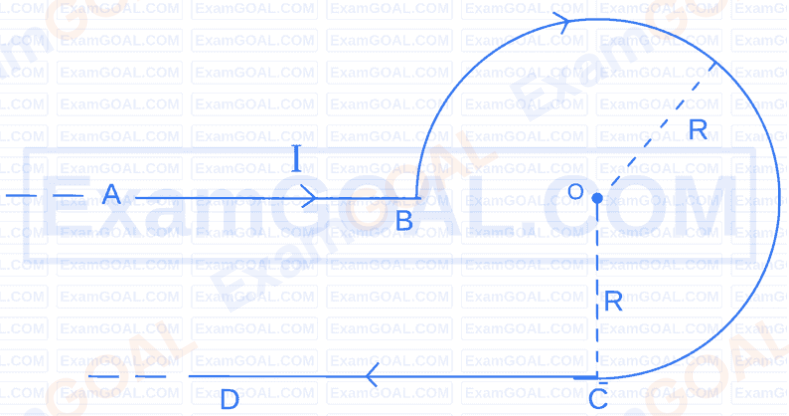
(A) $-8\,\hat{i}\ \mathrm{A\,m^2}$ (B) $\frac{8}{\pi}\,\hat{i}\ \mathrm{A\,m^2}$

(C) $-4\pi\,\hat{i}\ \mathrm{A\,m^2}$ (D) $4\pi\,\hat{i}\ \mathrm{A\,m^2}$

Q62. JEE Main 2022 (Online) 30th July Evening Shift

In the following loop, magnitude of the magnetic field produced at the centre (O) of the loop is : (Assume AB and CD are infinitely extended):

Open solution



(A) $\frac{\mu_0 I}{4\pi R}\left(\frac{3}{2}\pi\right)$ (B) $\frac{\mu_0 I}{2\pi R}$

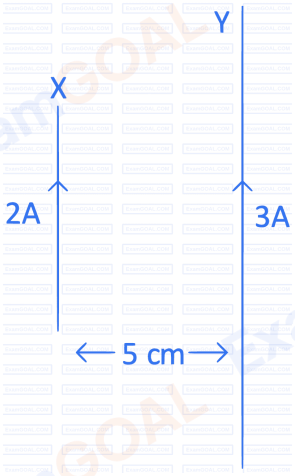
(C) $\frac{\mu_0 I}{4\pi R}\left(\frac{3}{2}\pi+1\right)$ (D) $\frac{\mu_0 I}{4\pi R}\left(\frac{3}{2}\pi-1\right)$

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Q63. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

A wire X of length $50\ \text{cm}$ carrying a current of $2\ \text{A}$ is placed parallel to a long wire Y of length $5\ \text{m}$. The wire Y carries a current of $3\ \text{A}$. The distance between two wires is $5\ \text{cm}$ and currents flow in the same direction. The force acting on the wire Y is



(A) $1.2 \times 10^{-5}\ \text{N}$ directed towards wire X.

(B) $1.2 \times 10^{-4}\ \text{N}$ directed away from wire X.

(C) $1.2 \times 10^{-4}\ \text{N}$ directed towards wire X.

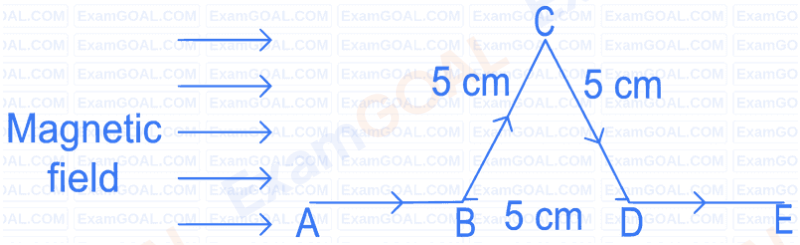
(D) $2.4 \times 10^{-5}\ \text{N}$ directed towards wire X.

Q64. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

A triangular shaped wire carrying $10\ \text{A}$ current is placed in a uniform magnetic field of $0.5\ \text{T}$, as shown in figure. The magnetic force on segment CD is

(Given $\text{BC} = \text{CD} = \text{BD} = 5\ \text{cm}$.)



(A) 0.126 N (B) 0.312 N (C) 0.216 N (D) 0.245 N

Q65. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

The magnetic field at the center of current carrying circular loop is B_1. The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2. The value of B_1/B_2 will be

(A) 9 : 4 (B) $12 : \sqrt{5}$ (C) 8 : 1 (D) $5 : \sqrt{3}$

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Q66. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

The current sensitivity of a galvanometer can be increased by :

(A) decreasing the number of turns

(B) increasing the magnetic field

(C) decreasing the area of the coil

(D) decreasing the torsional constant of the spring

Choose the most appropriate answer from the options given below :

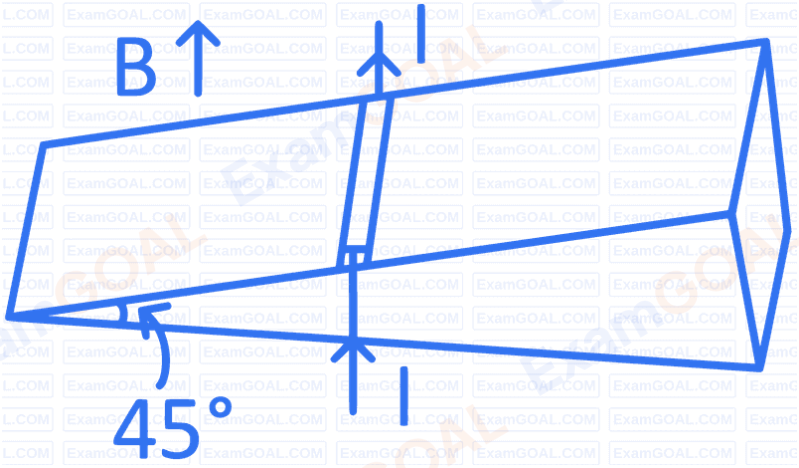
(A) (B) and (C) only **(B)** (C) and (D) only **(C)** (A) and (C) only **(D)** (B) and (D) only

Q67. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

As shown in the figure, a metallic rod of linear density $0.45\ \mathrm{kg\ m^{-1}}$ is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when $0.15\ \mathrm{T}$ magnetic field is acting on it in the vertical upward direction, will be :

{Use $g = 10\ \mathrm{m/s^2}$}



(A) 30 A **(B)** 15 A **(C)** 10 A **(D)** 3 A

Q68. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

A cyclotron is used to accelerate protons. If the operating magnetic field is $1.0\ \mathrm{T}$ and the radius of the cyclotron 'dees' is $60\ \mathrm{cm}$, the kinetic energy of the accelerated protons in MeV will be :

$[\text{use } \mathrm{m_p} = 1.6 \times 10^{-27}\ \mathrm{kg}, \mathrm{e} = 1.6 \times 10^{-19}\ \mathrm{C}\ ]$

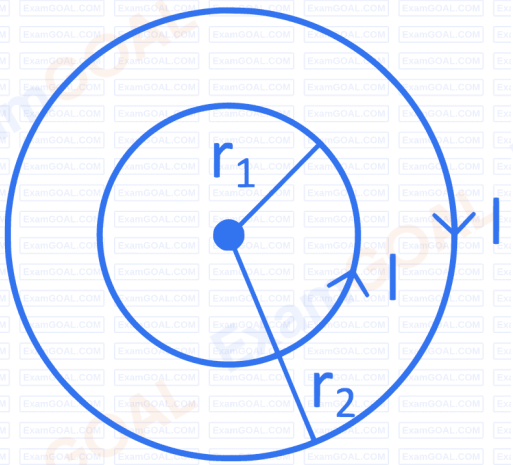
(A) 12 **(B)** 18 **(C)** 16 **(D)** 32

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Q69. JEE Main 2022 (Online) 26th July Evening Shift

Two concentric circular loops of radii $r_1 = 30\ \text{cm}$ and $r_2 = 50\ \text{cm}$ are placed in $\text{X} - \text{Y}$ plane as shown in the figure. A current $I = 7\ \text{A}$ is flowing through them in the direction as shown in figure. The net magnetic moment of this system of two circular loops is approximately :

Open solution



(A) $\frac{7}{2}\hat{k}\,\text{Am}^2$ (B) $-\frac{7}{2}\hat{k}\,\text{Am}^2$

(C) $7\,\hat{k}\,\text{Am}^2$ (D) $-7\,\hat{k}\,\text{Am}^2$

Q70. JEE Main 2022 (Online) 26th July Evening Shift

A velocity selector consists of electric field $\vec{E} = E\,\hat{k}$ and magnetic field $\vec{B} = B\,\hat{j}$ with $B = 12\,mT$. The value of E required for an electron of energy $728\,\text{eV}$ moving along the positive x-axis to pass undeflected is :

(Given, mass of electron $= 9.1 \times 10^{-31}\ \text{kg}$)

Open solution

(A) $192\,\text{kVm}^{-1}$ (B) $192\,\text{mVm}^{-1}$ (C) $9600\,\text{kVm}^{-1}$ (D) $16\,\text{kVm}^{-1}$

Q71. JEE Main 2022 (Online) 26th July Morning Shift

A charge particle is moving in a uniform magnetic field $(2\hat{i} + 3\hat{j})\ \text{T}$. If it has an acceleration of $(\alpha\hat{i} - 4\hat{j})\ \text{m/s}^2$, then the value of α will be :

Open solution

(A) 3 (B) 6 (C) 12 (D) 2

Q72. JEE Main 2022 (Online) 26th July Morning Shift

B_X and B_Y are the magnetic fields at the centre of two coils X and Y respectively each carrying equal current. If coil X has 200 turns and $20\ \text{cm}$ radius and coil Y has 400 turns and $20\ \text{cm}$ radius, the ratio of B_X and B_Y is :

Open solution

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1

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Q73. JEE Main 2022 (Online) 25th July Evening Shift

The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction B_2 at its centre. The ratio of $\frac{B_2}{B_1}$ is

Open solution

(A) $\frac{5}{2}$ (B) $\frac{25}{4}$ (C) $\frac{5}{4}$ (D) $\frac{25}{2}$

Q74. JEE Main 2022 (Online) 25th July Morning Shift

Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is $6:5$ and their respective masses ratio is $9:4$. Then, the ratio of their charges will be :

Open solution

(A) 8 : 5 (B) 5 : 4 (C) 5 : 3 (D) 8 : 7

Q75. JEE Main 2022 (Online) 30th June Morning Shift

A coil of n number of turns wound tightly in the form of a spiral with inner and outer radii r_1 and r_2 respectively. When a current of strength I is passed through the coil, the magnetic field at its centre will be :

Open solution

(A) $\frac{\mu_0 nI}{2(r_2-r_1)}$ (B) $\frac{\mu_0 nI}{r_2}$

(C) $\frac{\mu_0 nI}{r_2-r_1}\log_e\frac{r_1}{r_2}$ (D) $\frac{\mu_0 nI}{2(r_2-r_1)}\log_e\frac{r_2}{r_1}$

Q76. JEE Main 2022 (Online) 29th June Evening Shift

Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is 300 μT. The equal current flowing in the two conductors is :

Open solution

(A) 30A in the same direction. (B) 30A in the opposite direction.

(C) 60A in the opposite direction. (D) 300A in the opposite direction.

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Q77. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

Given below are two statements :

Statement I : The electric force changes the speed of the charged particle and hence changes its kinetic energy; whereas the magnetic force does not change the kinetic energy of the charged particle.

Statement II : The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q78. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

A charge particle moves along circular path in a uniform magnetic field in a cyclotron. The kinetic energy of the charge particle increases to 4 times its initial value. What will be the ratio of new radius to the original radius of circular path of the charge particle :

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 1 : 4

Q79. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

Two parallel, long wires are kept 0.20 m apart in vacuum, each carrying current of x A in the same direction. If the force of attraction per meter of each wire is 2×10^{-6} N, then the value of x is approximately :

(A) 1 (B) 2.4 (C) 1.4 (D) 2

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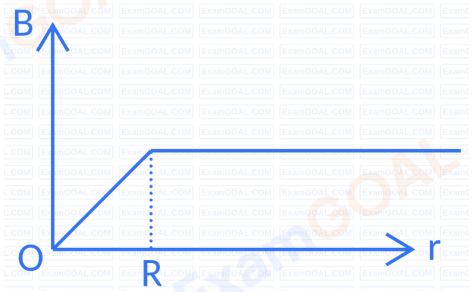
Q80. JEE Main 2022 (Online) 28th June Morning Shift

An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface.

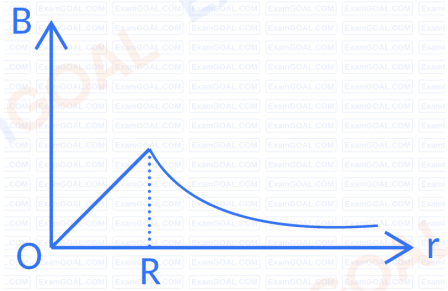
Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.

Open solution

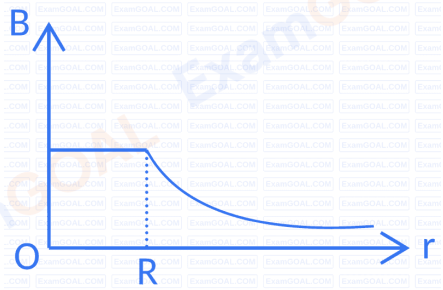
(A)



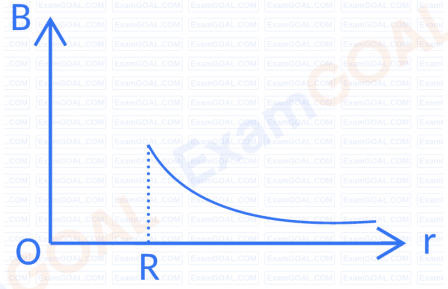
(B)



(C)



(D)



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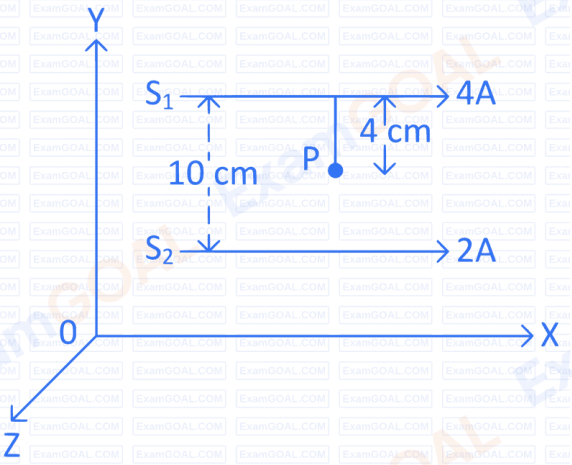
Q81. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

Two long parallel conductors S_1 and S_2 are separated by a distance 10 cm and carrying currents of 4A and 2A respectively. The conductors are placed along x-axis in X-Y plane. There is a point P located between the conductors (as shown in figure).

A charged particle of 3π coulomb is passing through the point P with velocity $\vec{v} = (2\hat{i} + 3\hat{j})$ m/s; where $\hat{i}$ and $\hat{j}$ represents unit vector along x & y axis respectively.

The force acting on the charged particle is $4\pi \times 10^{-5}(-x\hat{i} + 2\hat{j})$ N. The value of x is :



(A) 2 (B) 1 (C) 3 (D) -3

Q82. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

A proton and an alpha particle of the same velocity enter in a uniform magnetic field which is acting perpendicular to their direction of motion. The ratio of the radii of the circular paths described by the alpha particle and proton is :

(A) 1 : 4 (B) 4 : 1 (C) 2 : 1 (D) 1 : 2

Q83. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to

(A) B (B) 2B (C) 4B (D) $\frac{B}{2}$

Q84. JEE Main 2022 (Online) 25th June Morning Shift

Open solution

A long straight wire with a circular cross-section having radius R, is carrying a steady current I. The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance r (r < R) from its centre will be :

(A) $B \propto r^2$ (B) $B \propto r$ (C) $B \propto \frac{1}{r^2}$ (D) $B \propto \frac{1}{r}$

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Q85. JEE Main 2022 (Online) 24th June Evening Shift

A proton, a deuteron and an α-particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is :

(A) $1:\sqrt{2}:\sqrt{2}$ (B) $1:1:\sqrt{2}$ (C) $\sqrt{2}:1:1$ (D) $1:\sqrt{2}:1$

Open solution

Q86. JEE Main 2022 (Online) 24th June Morning Shift

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In an uniform magnetic field, speed and energy remains the same for a moving charged particle.

Reason (R) : Moving charged particle experiences magnetic force perpendicular to its direction of motion.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Open solution

Q87. JEE Main 2022 (Online) 24th June Morning Shift

The magnetic field at the centre of a circular coil of radius r, due to current I flowing through it, is B. The magnetic field at a point along the axis at a distance $\frac{r}{2}$ from the centre is :

(A) B/2 (B) 2B

(C) $\left(\frac{2}{\sqrt{5}}\right)^3 B$ (D) $\left(\frac{2}{\sqrt{3}}\right)^3 B$

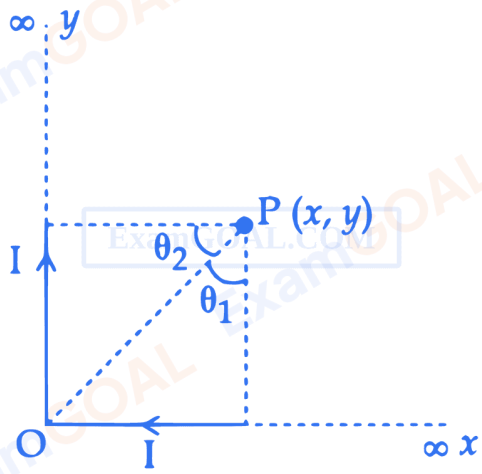
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Q88. JEE Main 2021 (Online) 1st September Evening Shift

Open solution

There are two infinitely long straight current carrying conductors and they are held at right angles to each other so that their common ends meet at the origin as shown in the figure given below. The ratio of current in both conductor is 1 : 1. The magnetic field at point P is _____.



(A) $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}+(x+y)\right]$

(B) $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}-(x+y)\right]$

(C) $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}-(x+y)\right]$

(D) $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}+(x+y)\right]$

Q89. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

A current of 1.5 A is flowing through a triangle, of side 9 cm each. The magnetic field at the centroid of the triangle is :

(Assume that the current is flowing in the clockwise direction.)

(A) 3×10^{-7} T, outside the plane of triangle

(B) $2\sqrt{3} \times 10^{-7}$ T, outside the plane of triangle

(C) $2\sqrt{3} \times 10^{-5}$ T, inside the plane of triangle

(D) 3×10^{-5} T, inside the plane of triangle

Q90. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

A coil having N turns is wound tightly in the form of a spiral with inner and outer radii 'a' and 'b' respectively. Find the magnetic field at centre, when a current I passes through coil:

(A) $\frac{\mu_0 IN}{2(b-a)}\log_e\left(\frac{b}{a}\right)$

(B) $\frac{\mu_0 I}{8}\left[\frac{a+b}{a-b}\right]$

(C) $\frac{\mu_0 I}{4(a-b)}\left[\frac{1}{a}-\frac{1}{b}\right]$

(D) $\frac{\mu_0 I}{8}\left(\frac{a-b}{a+b}\right)$

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Q91. JEE Main 2021 (Online) 27th August Evening Shift

A coaxial cable consists of an inner wire of radius 'a' surrounded by an outer shell of inner and outer radii 'b' and 'c' respectively. The inner wire carries an electric current i_0, which is distributed uniformly across cross-sectional area. The outer shell carries an equal current in opposite direction and distributed uniformly. What will be the ratio of the magnetic field at a distance x from the axis when (i) x < a and (ii) a < x < b ?

Open solution

(A) $\frac{x^2}{a^2}$ (B) $\frac{a^2}{x^2}$ (C) $\frac{x^2}{b^2-a^2}$ (D) $\frac{b^2-a^2}{x^2}$

Q92. JEE Main 2021 (Online) 27th August Morning Shift

Two ions of masses 4 amu and 16 amu have charges +2e and +3e respectively. These ions pass through the region of constant perpendicular magnetic field. The kinetic energy of both ions is same. Then :

Open solution

(A) lighter ion will be deflected less than heavier ion

(B) lighter ion will be deflected more than heavier ion

(C) both ions will be deflected equally

(D) no ion will be deflected

Q93. JEE Main 2021 (Online) 26th August Morning Shift

The fractional change in the magnetic field intensity at a distance 'r' from centre on the axis of current carrying coil of radius 'a' to the magnetic field intensity at the centre of the same coil is : (Take r < a)

Open solution

(A) $\frac{3}{2}\frac{a^2}{r^2}$ (B) $\frac{2}{3}\frac{a^2}{r^2}$

(C) $\frac{2}{3}\frac{r^2}{a^2}$ (D) $\frac{3}{2}\frac{r^2}{a^2}$

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Q94. JEE Main 2021 (Online) 27th July Evening Shift

Figures A and B show two long straight wires of circular cross-section (a and b with a < b), carrying current I which is uniformly distributed across the cross-section. The magnitude of magnetic field B varies with radius r and can be represented as :

Open solution

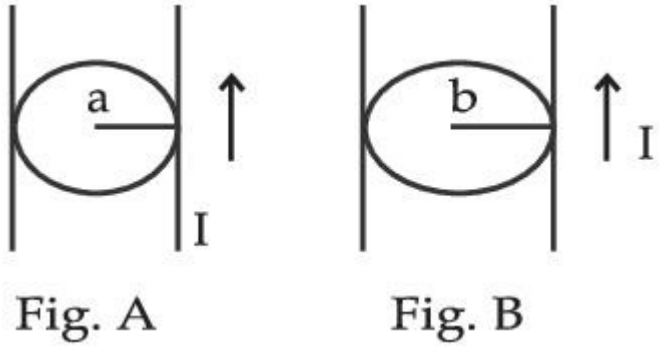
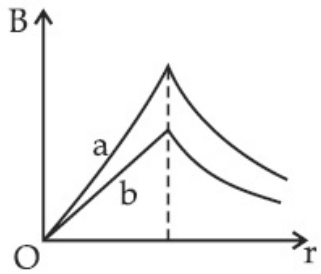
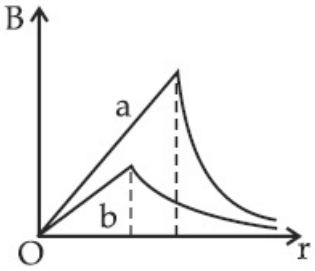


Fig. A Fig. B

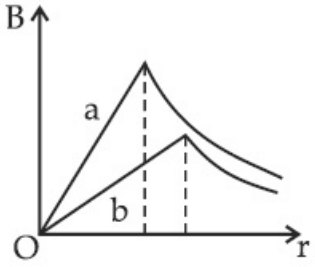
(A)



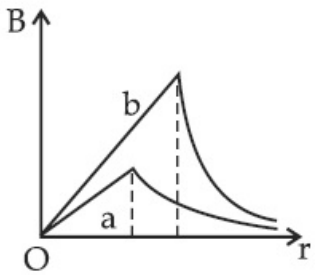
(B)



(C)



(D)



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Q95. JEE Main 2021 (Online) 25th July Evening Shift

Two ions having same mass have charges in the ratio 1 : 2. They are projected normally in a uniform magnetic field with their speeds in the ratio 2 : 3. The ratio of the radii of their circular trajectories is :

(A) 1 : 4 (B) 4 : 3 (C) 3 : 1 (D) 2 : 3

Open solution

Q96. JEE Main 2021 (Online) 20th July Morning Shift

A deuteron and an alpha particle having equal kinetic energy enter perpendicularly into a magnetic field. Let r_d and r_α be their respective radii of circular path. The value of $\frac{r_d}{r_\alpha}$ is equal to :

(A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

Open solution

Q97. JEE Main 2021 (Online) 18th March Evening Shift

A proton and an α-particle, having kinetic energies K_p and K_α respectively, enter into a magnetic field at right angles.

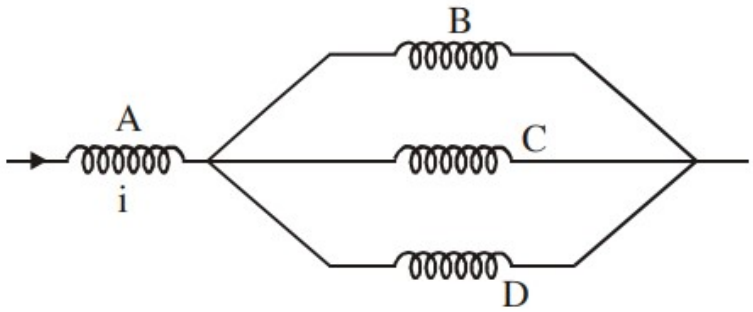
The ratio of the radii of trajectory of proton to that of α-particle is 2 : 1. The ratio of $K_p : K_\alpha$ is :

(A) 8 : 1 (B) 4 : 1 (C) 1 : 8 (D) 1 : 4

Open solution

Q98. JEE Main 2021 (Online) 18th March Morning Shift

Four identical long solenoids A, B, C and D are connected to each other as shown in the figure. If the magnetic field at the center of A is 3T, the field at the center of C would be : (Assume that the magnetic field is confined with in the volume of respective solenoid).



(A) 12T (B) 6T (C) 9T (D) 1T

Open solution

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Q99. JEE Main 2021 (Online) 18th March Morning Shift

A loop of flexible wire of irregular shape carrying current is placed in an external magnetic field. Identify the effect of the field on the wire.

(A) Loop assumes circular shape with its plane normal to the field.

(B) Loop assumes circular shape with its plane parallel to the field.

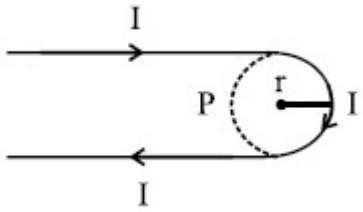
(C) Wire gets stretched to become straight.

(D) Shape of the loop remains unchanged.

Open solution

Q100. JEE Main 2021 (Online) 17th March Evening Shift

A hairpin like shape as shown in figure is made by bending a long current carrying wire. What is the magnitude of a magnetic field at point P which lies on the centre of the semicircle?



(A) $\frac{\mu_0 I}{4\pi r}(2-\pi)$

(B) $\frac{\mu_0 I}{2\pi r}(2-\pi)$

(C) $\frac{\mu_0 I}{4\pi r}(2+\pi)$

(D) $\frac{\mu_0 I}{2\pi r}(2+\pi)$

Open solution

Q101. JEE Main 2021 (Online) 17th March Morning Shift

A solenoid of 1000 turns per metre has a core with relative permeability 500. Insulated windings of the solenoid carry an electric current of 5A. The magnetic flux density produced by the solenoid is : (permeability of free space = $4\pi \times 10^{-7}$ H/m)

(A) πT

(B) $2 \times 10^{-3}\pi$ T

(C) $10^{-4}\pi$ T

(D) $\frac{\pi}{5}$ T

Open solution

Q102. JEE Main 2021 (Online) 16th March Evening Shift

A charge Q is moving $\vec{dl}$ distance in the magnetic field $\vec{B}$. Find the value of work done by $\vec{B}$.

(A) Zero

(B) -1

(C) Infinite

(D) 1

Open solution

Q103. JEE Main 2021 (Online) 25th February Morning Shift

A proton, a deuteron and an α particle are moving with same momentum in a uniform magnetic field. The ratio of magnetic forces acting on them is _____ and their speed is _____, in the ratio.

(A) 2 : 1 : 1 and 4 : 2 : 1

(B) 4 : 2 : 1 and 2 : 1 : 1

(C) 1 : 2 : 4 and 2 : 1 : 1

(D) 1 : 2 : 4 and 1 : 1 : 2

Open solution

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Q104. JEE Main 2021 (Online) 25th February Morning Shift

Magnetic fields at two points on the axis of a circular coil at a distance of 0.05 m and 0.2 m from the centre are in the ratio 8 : 1. The radius of coil is _____.

(A) 1.0 m (B) 0.15 m (C) 0.2 m (D) 0.1 m

Open solution

Q105. JEE Main 2020 (Online) 6th September Evening Slot

A charged particle going around in a circle can be considered to be a current loop. A particle of mass m carrying charge q is moving in a plane with speed v under the influence of magnetic field $\vec{B}$. The magnetic moment of this moving particle:

(A) $\frac{mv^2\vec{B}}{2B^2}$ (B) $-\frac{mv^2\vec{B}}{2B^2}$

(C) $-\frac{mv^2\vec{B}}{B^2}$ (D) $-\frac{mv^2\vec{B}}{2\pi B^2}$

Open solution

Q106. JEE Main 2020 (Online) 6th September Evening Slot

A square loop of side $2a$ and carrying current I is kept in xz plane with its centre at origin. A long wire carrying the same current I is placed parallel to z-axis and passing through point (0, b, 0), (b >> a). The magnitude of torque on the loop about z-axis will be :

(A) $\frac{2\mu_0 I^2 a^2}{\pi b}$ (B) $\frac{2\mu_0 I^2 a^2 b}{\pi(a^2+b^2)}$

(C) $\frac{\mu_0 I^2 a^2 b}{2\pi(a^2+b^2)}$ (D) $\frac{\mu_0 I^2 a^2}{2\pi b}$

Open solution

Q107. JEE Main 2020 (Online) 6th September Morning Slot

An electron is moving along +x direction with a velocity of 6×10^6 ms^{-1}. It enters a region of uniform electric field of 300 V/cm pointing along +y direction. The magnitude and direction of the magnetic field set up in this region such that the electron keeps moving along the x direction will be :

(A) 3×10^{-4} T, along −z direction (B) 5×10^{-3} T, along −z direction

(C) 5×10^{-3} T, along +z direction (D) 3×10^{-4} T, along +z direction

Open solution

Q108. JEE Main 2020 (Online) 6th September Morning Slot

A particle of charge q and mass m is moving with a velocity $-v\hat{i}$ (v $\neq$ 0) towards a large screen placed in the Y - Z plane at a distance d. If there is a magnetic field $\vec{B} = B_0\hat{k}$, the maximum value of v for which the particle will not hit the screen is :

(A) $\frac{2qdB_0}{m}$ (B) $\frac{qdB_0}{3m}$ (C) $\frac{qdB_0}{2m}$ (D) $\frac{qdB_0}{m}$

Open solution

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Q109. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

An iron rod of volume 10^{-3} m^3 and relative permeability 1000 is placed as core in a solenoid with 10 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment of the rod will be :

(A) 5×10^2 Am^2 (B) 0.5×10^2 Am^2 (C) 500×10^2 Am^2 (D) 50×10^2 Am^2

Q110. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

A square loop of side $2a$, and carrying current I, is kept in XZ plane with its centre at origin. A long wire carrying the same current I is placed parallel to the z-axis and passing through the point (0, b, 0), (b >> a). The magnitude of the torque on the loop about z-axis is given by :

(A) $\frac{2\mu_0 I^2 a^2}{\pi b}$ (B) $\frac{\mu_0 I^2 a^2}{2\pi b}$

(C) $\frac{\mu_0 I^2 a^3}{2\pi b^2}$ (D) $\frac{2\mu_0 I^2 a^3}{\pi b^2}$

Q111. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

A circular coil has moment of inertia 0.8 kg m^2 around any diameter and is carrying current to produce a magnetic moment of 20 Am^2 . The coil is kept initially in a vertical position and it can rotate freely around a horizontal diameter. When a uniform magnetic field of 4 T is applied along the vertical,it starts rotating around its horizontal diameter. The angular speed the coil acquires after rotating by 60° will be:

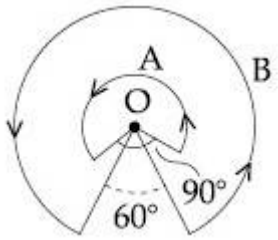
(A) $10\,\pi$ rad s^{-1} (B) $20\,\pi$ rad s^{-1}

(C) $10(3)^{1/4}$ rad s^{-1} (D) 20 rad s^{-1}

Q112. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

A wire A, bent in the shape of an arc of a circle, carrying a current of 2 A and having radius 2 cm and another wire B, also bent in the shape of arc of a circle, carrying a current of 3 A and having radius of 4 cm, are placed as shown in the figure. The ratio of the magnetic fields due to the wires A and B at the common centre O is :



(A) 4 : 6 (B) 6 : 4 (C) 2 : 5 (D) 6 : 5

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Q113. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

A charged particle carrying charge 1 μC is moving with velocity $\left(2\hat{i}+3\hat{j}+4\hat{k}\right)$ ms^{-1}. If an external magnetic field of $\left(5\hat{i}+3\hat{j}-6\hat{k}\right)\times 10^{-3}$ T exists in the region where the particle is moving then the force on the particle is $\vec{F}\times 10^{-9}$ N. The vector $\vec{F}$ is :

(A) $-0.30\hat{i}+0.32\hat{j}-0.09\hat{k}$

(B) $-300\hat{i}+320\hat{j}-90\hat{k}$

(C) $-30\hat{i}+32\hat{j}-9\hat{k}$

(D) $-3.0\hat{i}+3.2\hat{j}-0.9\hat{k}$

Q114. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

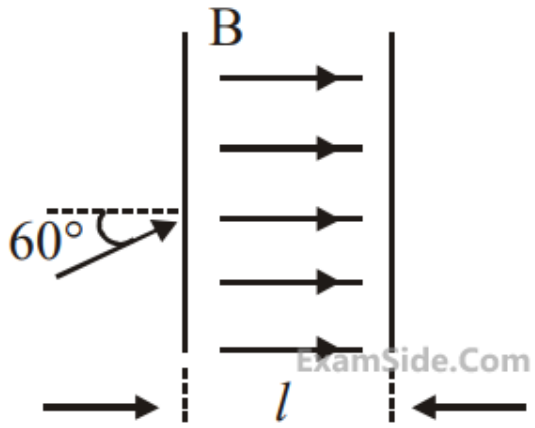
Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10 cm, 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is :

(A) $250\sqrt{3}$ (B) $5\sqrt{3}$ (C) $500\sqrt{3}$ (D) $50\sqrt{3}$

Q115. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

The figure shows a region of length 'l' with a uniform magnetic field of 0.3 T in it and a proton entering the region with velocity 4×10^5 ms^{-1} making an angle 60° with the field. If the proton completes 10 revolution by the time it cross the region shown, 'l' is close to
(mass of proton = 1.67×10^{-27} kg, charge of the proton = 1.6×10^{-19} C)



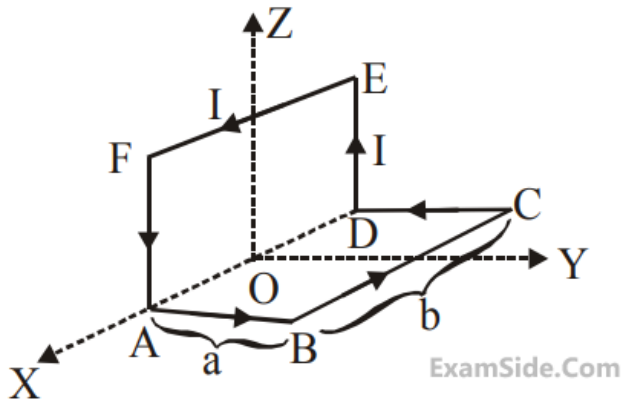
(A) 0.22 m (B) 0.11 m (C) 0.88 m (D) 0.44 m

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Q116. JEE Main 2020 (Online) 2nd September Evening Slot

A wire carrying current I is bent in the shape ABCDEFA as shown, where rectangle ABCDA and ADEFA are perpendicular to each other. If the sides of the rectangles are of lengths a and b, then the magnitude and direction of magnetic moment of the loop ABCDEFA is

Open solution



(A) $\sqrt{2}$abI, along $\left(\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}}\right)$

(B) abI, along $\left(\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}}\right)$

(C) $\sqrt{2}$abI, along $\left(\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}\right)$

(D) abI, along $\left(\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}\right)$

Q117. JEE Main 2020 (Online) 2nd September Morning Slot

A beam of protons with speed 4×10^5 ms^{-1} enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. The pitch of the resulting helical path of protons is close to :
(Mass of the proton = 1.67×10^{-27} kg, charge of the proton = 1.69×10^{-19} C)

Open solution

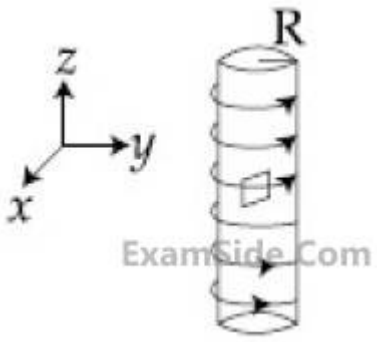
(A) 2 cm (B) 12 cm (C) 5 cm (D) 4 cm

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Q118. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I. The electron gun shoots an electron along the radius of the solenoid with speed v. If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard meaning) :



(A) $\frac{e\mu_0 nIR}{4m}$ (B) $\frac{e\mu_0 nIR}{m}$ (C) $\frac{e\mu_0 nIR}{2m}$ (D) $\frac{2e\mu_0 nIR}{m}$

Q119. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

A small circular loop of conducting wire has radius a and carries current I. It is placed in a uniform magnetic field B perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T. If the mass of the loop is m then :

(A) $T = \sqrt{\frac{2m}{IB}}$ (B) $T = \sqrt{\frac{\pi m}{IB}}$

(C) $T = \sqrt{\frac{\pi m}{2IB}}$ (D) $T = \sqrt{\frac{2\pi m}{IB}}$

Q120. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and $2a$, respectively from the axis of the wire is :

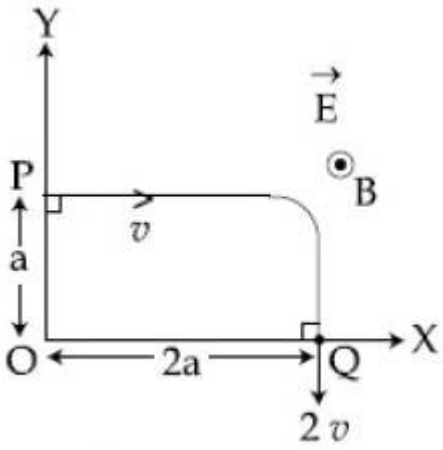
(A) 2 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

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Q121. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

A charged particle of mass 'm' and charge 'q' moving under the influence of uniform electric field $\vec{Ei}$ and a uniform magnetic field $\vec{Bk}$ follows a trajectory from point P to Q as shown in figure. The velocities at P and Q are respectively, $\vec{vi}$ and $-2\vec{vj}$. Then which of the following statements (A, B, C, D) are correct ?
(Trajectory shown is schematic and not to scale) :



(A) E = $\frac{3}{4}\left(\frac{mv^2}{qa}\right)$

(B) Rate of work done by the electric field at P is $\frac{3}{4}\left(\frac{mv^3}{a}\right)$

(C) Rate of work done by both the fields at Q is zero

(D) The difference between the magnitude of angular momentum of the particle at P and Q is 2mav.

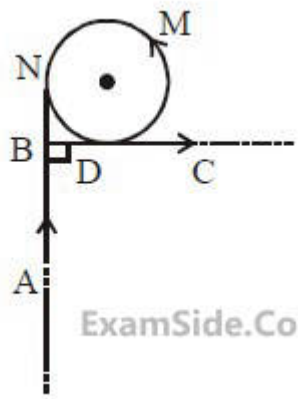
(A) (A), (B), (C), (D)
(B) (A), (B), (C)
(C) (A), (C), (D)
(D) (B), (C), (D)

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Q122. JEE Main 2020 (Online) 8th January Evening Slot

A very long wire ABDMNDC is shown in figure carrying current I. AB and BC parts are straight, long and at right angle. At D wire forms a circular turn DMND of radius R. AB, BC parts are tangential to circular turn at N and D. Magnetic field at the centre of circle is :

Open solution



(A) $\frac{\mu_0 I}{2\pi R}(\pi+1)$

(B) $\frac{\mu_0 I}{2\pi R}\left(\pi-\frac{1}{\sqrt{2}}\right)$

(C) $\frac{\mu_0 I}{2R}$

(D) $\frac{\mu_0 I}{2\pi R}\left(\pi+\frac{1}{\sqrt{2}}\right)$

Q123. JEE Main 2020 (Online) 8th January Morning Slot

Proton with kinetic energy of 1MeV moves from south to north. It gets an acceleration of 10^{12} m/s^2 by an applied magnetic field (west to east). The value of magnetic field : (Rest mass of proton is 1.6 × 10^{-27} kg) :

Open solution

(A) 0.71mT (B) 7.1mT (C) 0.071mT (D) 71mT

Q124. JEE Main 2020 (Online) 7th January Evening Slot

A particle of mass m and charge q has an initial velocity $\vec{v}=v_0\hat{j}$. If an electric field $\vec{E}=E_0\hat{i}$ and magnetic field $\vec{B}=B_0\hat{i}$ act on the particle, its speed will double after a time:

Open solution

(A) $\frac{3mv_0}{qE_0}$

(B) $\frac{\sqrt{2}mv_0}{qE_0}$

(C) $\frac{\sqrt{3}mv_0}{qE_0}$

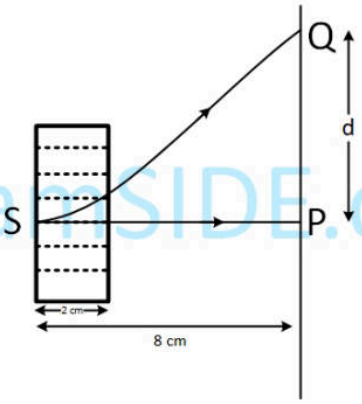
(D) $\frac{2mv_0}{qE_0}$

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Q125. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

An electron, moving along the x-axis with an initial energy of 100 eV, enters a region of magnetic field $\vec{B} = (1.5 \times 10^{-3} T)\hat{k}$ at S (See figure). The field extends between x = 0 and x = 2 cm. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance d between P and Q (on the screen) is : (electron's charge = 1.6×10^{-19} C, mass of electron = 9.1×10^{-31} kg)

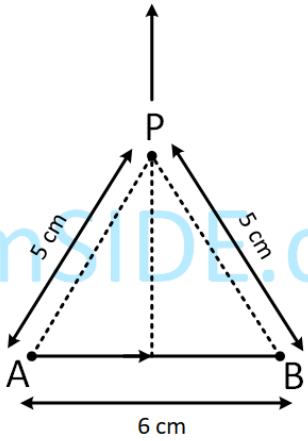


(A) 2.25 cm (B) 12.87 cm (C) 1.22 cm (D) 11.65 cm

Q126. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

Find the magnetic field at point P due to a straight line segment AB of length 6 cm carrying a current of 5A. (See figure) ($\mu_0 = 4\pi \times 10^{-7}$ N-A^{-2})



(A) 1.5×10^{-5} T (B) 3.0×10^{-5} T (C) 2.0×10^{-5} T (D) 2.5×10^{-5} T

Q127. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

A thin ring of 10 cm radius carries a uniformly distributed charge. The ring rotates at a constant angular speed of 40 π rad s^{-1} about its axis, perpendicular to its plane. If the magnetic field at its centre is 3.8×10^{-9} T, then the charge carried by the ring is close to ($\mu_0 = 4\pi \times 10^{-7}$ N/A^2).

(A) 7×10^{-6} C (B) 4×10^{-5} C (C) 2×10^{-6} C (D) 3×10^{-5} C

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Q128. JEE Main 2019 (Online) 10th April Evening Slot

A square loop is carrying a steady current I and the magnitude of its magnetic dipole moment is m. if this square loop is changed to a circular loop and it carries the same current, the magnitude of the magnetic dipole moment of circular loop will be:

Open solution

(A) $\frac{m}{\pi}$ (B) $\frac{3m}{\pi}$ (C) $\frac{2m}{\pi}$ (D) $\frac{4m}{\pi}$

Q129. JEE Main 2019 (Online) 10th April Evening Slot

The magnitude of the magnetic field at the centre of an equilateral triangular loop of side 1 m which is carrying a current of 10 A is :

[Take $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}]

Open solution

(A) 3 μT (B) 18 μT (C) 9 μT (D) 1 μT

Q130. JEE Main 2019 (Online) 10th April Morning Slot

A proton, an electron, and a Helium nucleus, have the same energy. They are in circular orbits in a plane due to magnetic field perpendicualr to the plane. Let r_p, r_e and r_{He} be their respective radii, then

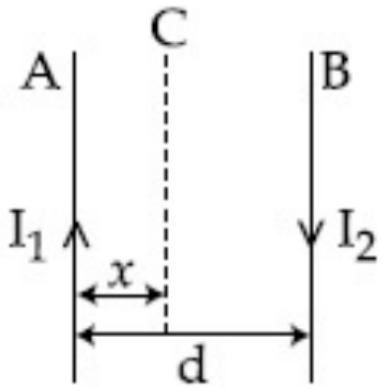
Open solution

(A) $r_e < r_p < r_{He}$ (B) $r_e < r_p = r_{He}$ (C) $r_e > r_p > r_{He}$ (D) $r_e > r_p = r_{He}$

Q131. JEE Main 2019 (Online) 10th April Morning Slot

Two wires A & B are carrying currents I_1 & I_2 as shown in the figure. The separation between them is d. A third wire C carrying a current I is to be kept parallel to them at a distance x from A such that the net force acting on it is zero. The possible values of x are :

Open solution



(A) $x = \left(\frac{I_1}{I_1+I_2}\right)d \; and \; x = \left(\frac{I_2}{I_1-I_2}\right)d$

(B) $x = \left(\frac{I_2}{I_1+I_2}\right)d \; and \; x = \left(\frac{I_2}{I_1-I_2}\right)d$

(C) $x = \left(\frac{I_1}{I_1-I_2}\right)d \; and \; x = \left(\frac{I_2}{I_1+I_2}\right)d$

(D) $x = \pm\frac{I_1 d}{I_1-I_2}$

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Q132. JEE Main 2019 (Online) 9th April Evening Slot

A moving coil galvanometer has a coil with 175 turns and area 1 cm^2. It uses a torsion band of torsion constant 10^{-6} N-m/rad. The coil is placed in a maganetic field B parallel to its plane. The coil deflects by 1° for a current of 1 mA. The value of B (in Tesla) is approximately :-

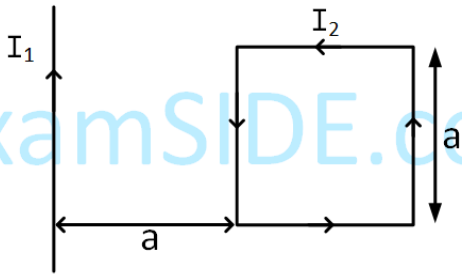
Open solution

(A) 10^{-4} (B) 10^{-2} (C) 10^{-1} (D) 10^{-3}

Q133. JEE Main 2019 (Online) 9th April Morning Slot

A rigid square loop of side 'a' and carrying current I_2 is lying on a horizontal surface near a long current I_1 carrying wire in the same plane as shown in figure. The net force on the loop due to wire will be :

Open solution



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(A) Repulsive and equal to $\mu_0 I_1 I_2/4\pi$ (B) Attractive and equal to $\mu_0 I_1 I_2/3\pi$

(C) Repulsive and equal to $\mu_0 I_1 I_2/2\pi$ (D) Zero

Q134. JEE Main 2019 (Online) 9th April Morning Slot

A rectangular coil (Dimension 5 cm × 2.5 cm) with 100 turns, carrying a current of 3 A in the clock-wise direction is kept centered at the origin and in the X-Z plane. A magnetic field of 1 T is applied along X-axis. If the coil is tilted through 45° about Z-axis, then the torque on the coil is :

Open solution

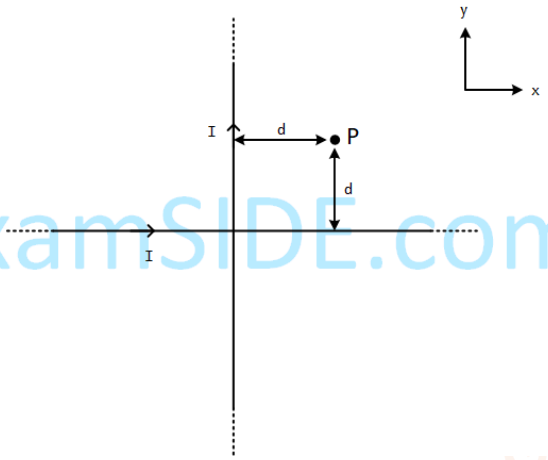
(A) 0.42 Nm (B) 0.55 Nm (C) 0.38 Nm (D) 0.27 Nm

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Q135. JEE Main 2019 (Online) 8th April Evening Slot

Two very long, straight, and insulated wires are kept at 90° angle from each other in xy-plane as shown in the figure. These wires carry currents of equal magnitude I, whose directions are shown in the figure. The net magnetic field at point P will be :

Open solution



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(A) $\frac{+\mu_0 I}{\pi d}\left(\hat{z}\right)$

(B) $-\frac{\mu_0 I}{2\pi d}\left(\hat{x}+\hat{y}\right)$

(C) Zero

(D) $\frac{\mu_0 I}{2\pi d}\left(\hat{x}+\hat{y}\right)$

Q136. JEE Main 2019 (Online) 8th April Morning Slot

A circular coil having N turns and radius r carries a current I. It is held in the XZ plane in a magnetic field $B\hat{i}$. The torque on the coil due to the magnetic field is :

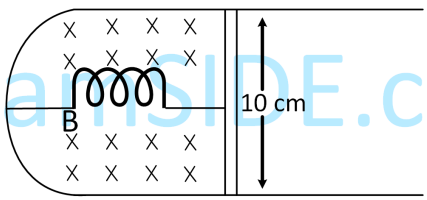
Open solution

(A) $\frac{Br^2 I}{\pi N}$ (B) $B\pi r^2 IN$ (C) Zero (D) $\frac{B\pi r^2 I}{N}$

Q137. JEE Main 2019 (Online) 8th April Morning Slot

A thin strip 10 cm long is on a U shaped wire of negligible resistance and it is connected to a spring of spring constant 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T. If the strip is pulled from its equilibrium position and released, the number of oscillations it performs before its amplitude decreases by a factor of e is N. If the mass of the strip is 50 grams, its resistance 10 Ω and air drag negligible, N will be close to :

Open solution



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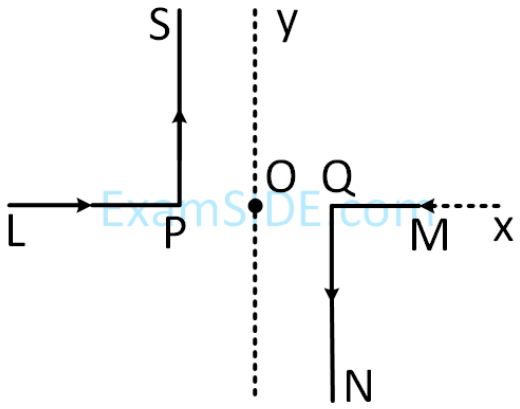
(A) 50000 (B) 1000 (C) 5000 (D) 10000

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Q138. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

As shown in the figure, two infinitely long, identical wires are bent by 90° and placed in such a way that the segments LP and QM are along the x-axis, while segments PS and QN are parallel to the y-axis. If OP = OQ = 4cm, and the magnitude of the magnetic field at O is 10^{-4} T, and the two wires carry equal currents (see figure), the magnitude of the current in each wire and the direction of the magnetic field at O will be ($\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}) :



(A) 40 A, perpendicular into the page

(B) 40 A, perpendicular out of the page

(C) 20 A, perpendicular into the page

(D) 20 A, perpendicular out of the page

Q139. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

A proton and an α-particle (with their masses in the ratio of 1 : 4 and charges in the ratio of 1 : 2) are accelerated from rest through a potential difference V. If a uniform magnetic field (B) is set up perpendicular to their velocities, the ratio of the radii $r_p : r_\alpha$ of the circular paths described by them will be ;

(A) $1 : \sqrt{3}$ (B) 1 : 3 (C) $1 : \sqrt{2}$ (D) 1 : 2

Q140. JEE Main 2019 (Online) 11th January Evening Slot

Open solution

The region between y = 0 and y = d contains a magnetic field $\vec{B} = B\hat{z}$. A particle of mass m and charge q enters the region with a velocity $\vec{v} = v\hat{i}$. If $d = \frac{mv}{2qB}$, the acceleration of the charged particle at the point of its emergence at the other side is :

(A) $\frac{qvB}{m}\left(-\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}\right)$

(B) $\frac{qvB}{m}\left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)$

(C) $\frac{qvB}{m}\left(\frac{-\hat{j}+\hat{i}}{\sqrt{2}}\right)$

(D) $\frac{qvB}{m}\left(\frac{\hat{j}+\hat{i}}{\sqrt{2}}\right)$

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Q141. JEE Main 2019 (Online) 11th January Evening Slot

A particle of mass m and charge q is in an electric and magnetic field given by $\overrightarrow{E} = 2\hat{i} + 3\hat{j};\ \overrightarrow{B} = 4\hat{j} + 6\hat{k}$.

Open solution

The charged particle is shifted from the origin to the point P(x = 1; y = 1) along a straight path. The magnitude of the total work done is :

(A) (2.5) q (B) (0.35) q (C) (0.15) q (D) 5 q

Q142. JEE Main 2019 (Online) 11th January Morning Slot

In an experiment, electrons are accelerated, from rest, by applying a voltage of 500 V. Calculate the radius of the path if a magnetic field 100 mT is then applied. [Charge of the electron = 1.6×10^{-19} C Mass of the electron = 9.1×10^{-31} kg]

Open solution

(A) 7.5×10^{-4} m (B) 7.5×10^{-3} m (C) 7.5 m (D) 7.5×10^{-2} m

Q143. JEE Main 2019 (Online) 10th January Evening Slot

A hoop and a solid cylinder of same mass and radius are made of a permanent magnetic material with their magnetic moment parallel to their respective axes. But the magnetic moment of hoop is twice of solid cylinder. They are placed in a uniform magnetic field in such a manner that their magnetic moments make a small angle with the field. If the oscillation periods of hoop and cylinder are T_h and T_c respectively, then -

Open solution

(A) $T_h = 1.5\ T_c$ (B) $T_h = T_c$ (C) $T_h = 2T_c$ (D) $T_h = 0.5\ T_c$

Q144. JEE Main 2019 (Online) 10th January Morning Slot

An insulating thin rod of length l has a linear charge density $\rho(x) = \rho_0 \frac{x}{l}$ on it. The rod is rotated about an axis passing through the origin (x = 0) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is -

Open solution

(A) $\frac{\pi}{3} n\rho l^3$ (B) $\frac{\pi}{4} n\rho l^3$ (C) $n\rho l^3$ (D) $\pi n\rho l^3$

Q145. JEE Main 2019 (Online) 9th January Evening Slot

A particle having the same charge as of electron moves in a circular path of radius 0.5 cm under the influence of a magnetic field of 0.5 T. If an electric field of 100 V/m makes it to move in a straight path, then the mass of the particle is (Given charge of electron = 1.6×10^{-19} C)

Open solution

(A) 9.1×10^{-31} kg (B) 1.6×10^{-27} kg

(C) 1.6×10^{-19} kg (D) 2.0×10^{-24} kg

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Q146. JEE Main 2019 (Online) 9th January Evening Slot

One of the two identical conducting wires of length L is bent in the form of a circular loop and the other one into a circular coil of N identical turns. If the same current is passed in both, the ratio of the magnetic field at the centre of the loop (B_L) to that at the center of the coil (B_C), i.e. $\frac{B_L}{B_C}$ will be :

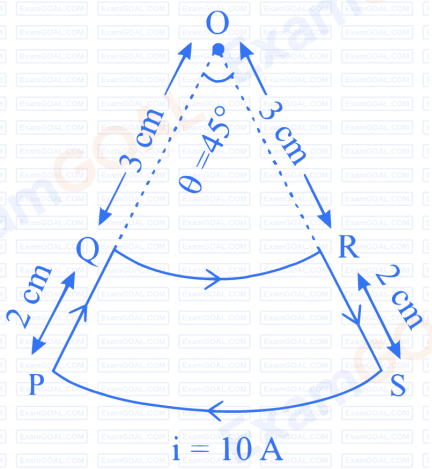
Open solution

(A) N (B) $\frac{1}{N}$ (C) N^2 (D) $\frac{1}{N^2}$

Q147. JEE Main 2019 (Online) 9th January Morning Slot

A current loop, having two circular arcs joined by two radial lines is shown in the figure. It carries a current of 10 A. The magnetic field at point O will be close to :

Open solution



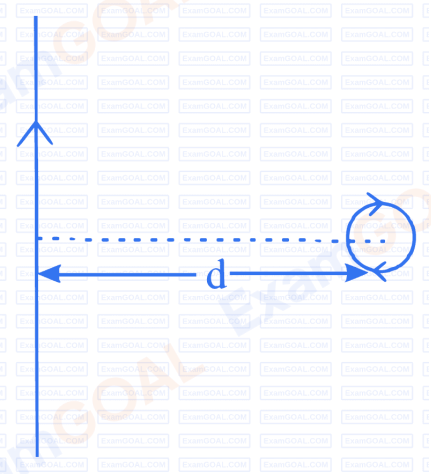
(A) 1.0×10^{-7} T (B) 1.5×10^{-7} T (C) 1.5×10^{-5} T (D) 1.0×10^{-5} T

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Q148. JEE Main 2019 (Online) 9th January Morning Slot

An infinitely long current carrying wire and a small current carrying loop are in the plane of the paper as shown. The radius of the loop is a and distance of its centre from the wire is d (d > > a). If the loop applies a force F on the wire then :

Open solution



(A) $F = 0$

(B) $F \propto \left(\frac{a}{d}\right)$

(C) $F \propto \left(\frac{a^2}{d^3}\right)$

(D) $F \propto \left(\frac{a}{d}\right)^2$

Q149. JEE Main 2018 (Online) 16th April Morning Slot

A charge q is spread uniformly over an insulated loop of radius r. If it is rotated with an angular velocity ω with resect to normal axis then the magnetic moment of the loop is :

Open solution

(A) $q\,\omega r^2$

(B) $\frac{4}{3}\,q\,\omega r^2$

(C) $\frac{3}{2}\,q\,\omega r^2$

(D) $\frac{1}{2}\,q\,\omega r^2$

Q150. JEE Main 2018 (Offline)

An electron, a proton and an alpha particle having the same kinetic energy are moving in circular orbits of radii r_e, r_p, r_α respectively in a uniform magnetic field B. The relation between r_e, r_p, r_α is:

Open solution

(A) $r_e < r_\alpha < r_p$

(B) $r_e > r_p = r_\alpha$

(C) $r_e < r_p = r_\alpha$

(D) $r_e < r_p < r_\alpha$

Q151. JEE Main 2018 (Offline)

The dipole moment of a circular loop carrying a current I, is m and the magnetic field at the centre of the loop is B_1. When the dipole moment is doubled by keeping the current constant, the magnetic field at the centre of the loop is B_2. The ratio $\frac{B_1}{B_2}$ is:

Open solution

(A) 2

(B) $\sqrt{3}$

(C) $\sqrt{2}$

(D) $\frac{1}{\sqrt{2}}$

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Q152. JEE Main 2018 (Online) 15th April Evening Slot

A current of 1 A is flowing on the sides of an equilateral triangle of side 4.5×10^{-2} m. The magnetic field at the center of the triangle will be :

Open solution

(A) 2×10^{-5} Wb/m^2

(B) Zero

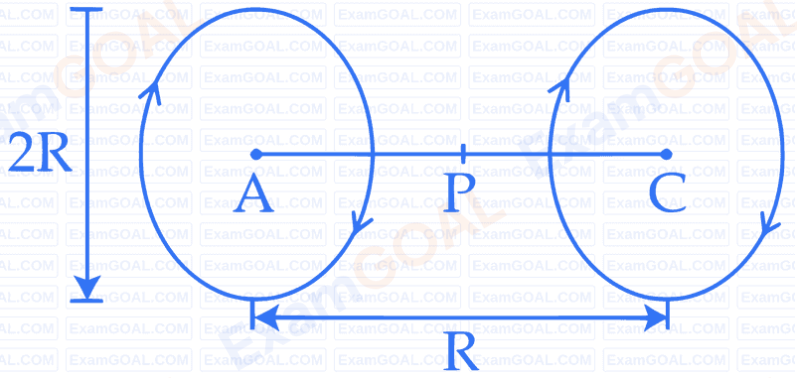
(C) 8×10^{-5} Wb/m^2

(D) 4×10^{-5} Wb/m^2

Q153. JEE Main 2018 (Online) 15th April Morning Slot

A Helmholtz coil has a pair of loops, each with N turns and radius R. They are placed coaxially at distance R and the same current I flows through the loops in the same direction. The magnitude of the magnetic field at P, midway between the centers A and C, is given by [Refer to figure given below] :

Open solution



(A) $\frac{8N\mu_0 I}{5^{1/2}R}$

(B) $\frac{8N\mu_0 I}{5^{3/2}R}$

(C) $\frac{4N\mu_0 I}{5^{1/2}R}$

(D) $\frac{4N\mu_0 I}{5^{3/2}R}$

Q154. JEE Main 2017 (Online) 9th April Morning Slot

A negative test charge is moving near a long straight wire carrying a current. The force acting on the test charge is parallel to the direction of the current. The motion of the charge is :

Open solution

(A) away from the wire

(B) towards the wire

(C) parallel to the wire along the current

(D) parallel to the wire opposite to the current

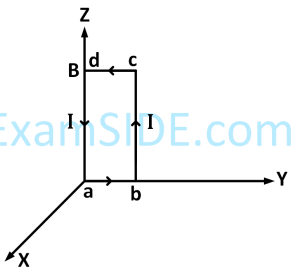
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Q155. JEE Main 2017 (Online) 9th April Morning Slot

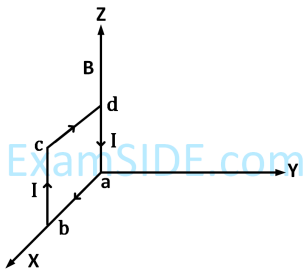
A uniform magnetic field B of 0.3 T is along the positive Z-direction. A rectangular loop (abcd) of sides 10 cm × 5 cm carries a current I of 12 A. Out of the following different orientations which one corresponds to stable equilibrium ?

Open solution

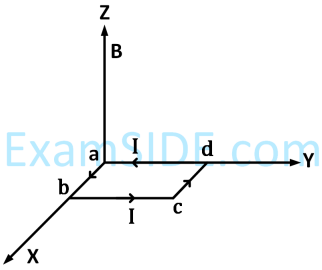
(A)



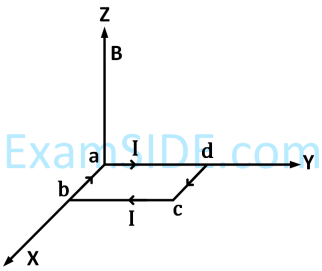
(B)



(C)



(D)



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Q156. JEE Main 2017 (Online) 8th April Morning Slot

In a certain region static electric and magnetic fields exist. The magnetic field is given by $\vec{B} = B_0\left(\hat{i} + 2\hat{j} - 4\hat{k}\right)$. If a test charge moving with a velocity $\vec{v} = v_0\left(3\hat{i} - \hat{j} + 2\hat{k}\right)$ experiences no force in that region, then the electric field in the region, in SI units, is :

Open solution

(A) $\vec{E} = -v_0 B_0\left(3\hat{i} - 2\hat{j} - 4\hat{k}\right)$

(B) $\vec{E} = -v_0 B_0\left(\hat{i} + \hat{j} + 7\hat{k}\right)$

(C) $\vec{E} = v_0 B_0\left(14\hat{j} + 7\hat{k}\right)$

(D) $\vec{E} = -v_0 B_0\left(14\hat{j} + 7\hat{k}\right)$

Q157. JEE Main 2017 (Online) 8th April Morning Slot

A magnetic dipole in a constant magnetic field has :

Open solution

(A) maximum potential energy when the torque is maximum.

(B) zero potential energy when the torque is minimum.

(C) zero potential energy when the torque is maximum.

(D) minimum potential energy when the torque is maximum.

Q158. JEE Main 2016 (Online) 9th April Morning Slot

A magnetic dipole is acted upon by two magnetic fields which are inclined to each other at an angle of 75°. One of the fields has a magnitude of 15 mT. The dipole attains stable equilibrium at an angle of 30° with this field. The magnitude of the other field (in mT) is close to

Open solution

(A) 11 (B) 36 (C) 1 (D) 1060

Q159. JEE Main 2016 (Online) 9th April Morning Slot

To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ. If a shunt of resistance S is needed to get half deflection then G, R and S are related by the equation :

Open solution

(A) 2S (R + G) = RG (B) S (R + G) = RG (C) 2S = G (D) 2G = S

Q160. JEE Main 2016 (Offline)

Two identical wires A and B, each of length 'l', carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side 'a'. If B_A and B_B are the values of magnetic fields at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is:

Open solution

(A) $\frac{\pi^2}{16}$ (B) $\frac{\pi^2}{8\sqrt{2}}$ (C) $\frac{\pi^2}{8}$ (D) $\frac{\pi^2}{16\sqrt{2}}$

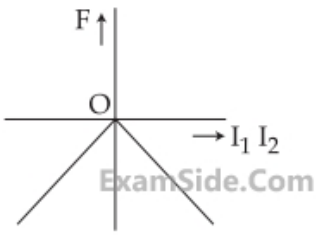
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Q161. JEE Main 2015 (Online) 11th April Morning Slot

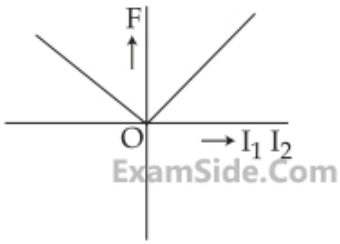
Open solution

Two long straight parallel wires, carrying (adjustable) currents I_1 and I_2, are kept at a distance d apart. If the force 'F' between the two wires is taken as 'positive' when the wires repel each other and 'negative' when the wires attract each other, the graph showing the dependence of 'F' , on the product I_1I_2, would be :

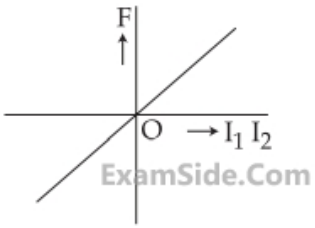
(A)



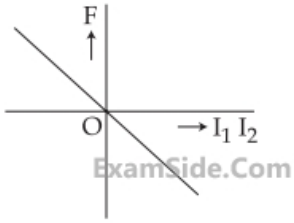
(B)



(C)



(D)

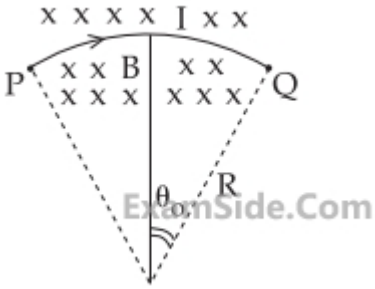


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Q162. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

A wire carrying current I is tied between points P and Q and is in the shape of a circular arch of radius R due to a uniform magnetic field B (perpendicular to the plane of the paper, shown by xxx) in the vicinity of the wire. If the wire subtends an angle $2\theta_0$ at the center of the circle (of which it forms an arch) then the tension in the wire is :



(A) IBR

(B) $\frac{IBR}{\sin\theta_0}$

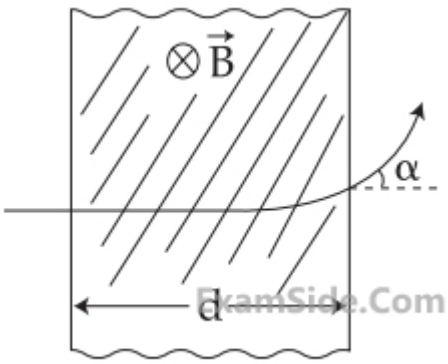
(C) $\frac{IBR}{2\sin\theta_0}$

(D) $\frac{IBR\theta_0}{\sin\theta_0}$

Q163. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field B. The field occupies a region of space by width 'd'. If 'α' be the angle of deviation of proton from initial direction of motion (see figure), the value of sin α will be :



(A) $\frac{B}{2}\sqrt{\frac{qd}{mV}}$

(B) $\frac{B}{d}\sqrt{\frac{q}{2mV}}$

(C) $Bd\sqrt{\frac{q}{2mV}}$

(D) $qV\sqrt{\frac{Bd}{2m}}$

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Q164. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

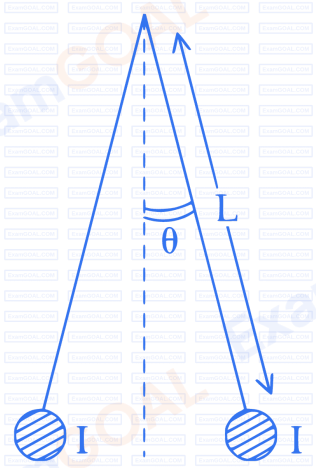
A 25 cm long solenoid has radius 2 cm and 500 total number of turns. It carries a current of 15 A. If it is equivalent to a magnet of the same size and magnetization $\overrightarrow{M}$ (magnetic moment/volume), then $|\overrightarrow{M}|$ is :

(A) 3π Am^{-1} (B) 30000 Am^{-1} (C) 300 Am^{-1} (D) 30000 π Am^{-1}

Q165. JEE Main 2015 (Offline)

Open solution

Two long current carrying thin wires, both with current I, are held by insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle $'\theta'$ with the vertical. If wires have mass λ per unit-length then the value of I is :
(g = gravitational acceleration)



(A) $2\sqrt{\frac{\pi gL}{\mu_0}\tan\theta}$ (B) $\sqrt{\frac{\pi\lambda gL}{\mu_0}\tan\theta}$

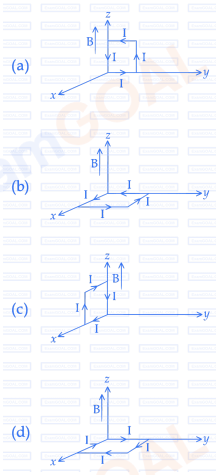
(C) $\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$ (D) $2\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$

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Q166. JEE Main 2015 (Offline)

Open solution

A rectangular loop of sides $10\,cm$ and $5\,cm$ carrying a current I of $12A$ is placed in different orientations as shown in the figures below :



If there is a uniform magnetic field of $0.3\,T$ in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium ?

(A) (B) and (D), respectively

(B) (B) and (C), respectively

(C) (A) and (B), respectively

(D) (A) and (C), respectively

Q167. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

A positive charge 'q' of mass 'm' is moving along the + x axis. We wish to apply a uniform magnetic field B for time Δt so that the charge reverses its direction crossing the y axis at a distance d. Then :

(A) B = $\frac{mv}{qd}$ and $\Delta t = \frac{\pi d}{v}$

(B) B = $\frac{mv}{2qd}$ and $\Delta t = \frac{\pi d}{2v}$

(C) B= $\frac{2mv}{qd}$ and $\Delta t = \frac{\pi d}{2v}$

(D) B = $\frac{2mv}{qd}$ and $\Delta t = \frac{\pi d}{v}$

Q168. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

Consider two thin identical conducting wires covered with very thin insulating material. One of the wires is bent into a loop and produces magnetic field B_1, at its center when a current I passes through it. The second wire is bent into a coil with three identical loops adjacent to each other and produces magnetic field B_2 at the centre of the loops when current I/3 passes through it. The ratio $B_1 : B_2$ is :

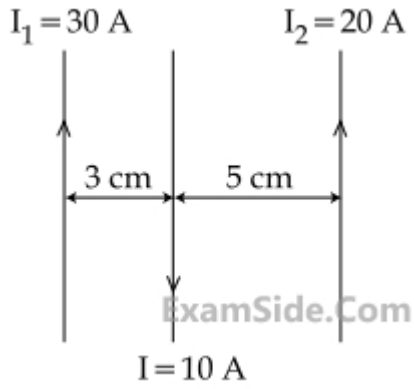
(A) 1 : 1 **(B)** 1 : 3 **(C)** 1 : 9 **(D)** 9 : 1

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Q169. JEE Main 2014 (Online) 11th April Morning Slot

Three straight parallel current carrying conductors are shown in the figure. The force experienced by the middle conductor of length 25 cm is :

Open solution



ExamSide.Com

(A) 3×10^{-4} N toward right

(B) 6×10^{-4} N toward left

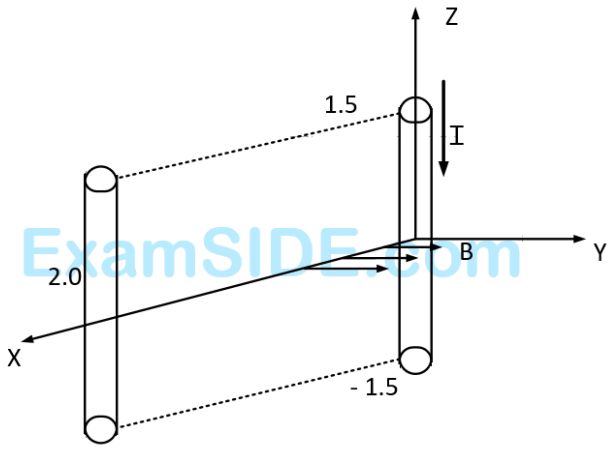
(C) 9×10^{-4} N toward left

(D) Zero

Q170. JEE Main 2014 (Offline)

A conductor lies along the z-axis at $-1.5 \leq z < 1.5\, m$ and carries a fixed current of $10.0\ A$ in $-\hat{a}_z$ direction (see figure). For a field $\vec{B} = 3.0 \times 10^{-4}\, e^{-0.2x}\ \hat{a}_y\ T$, find the power required to move the conductor at constant speed to $x = 2.0\ m$, $y = 0\ m$ in $5 \times 10^{-3} s$. Assume parallel motion along the x-axis.

Open solution



ExamSIDE.com

(A) $1.57W$

(B) $2.97W$

(C) $14.85\ W$

(D) $29.7W$

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Q171. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

One of the two small circular coils, (none of them having any self-inductance) is suspended with a V-shaped copper wire, with plane horizontal. The other coil is placed just below the first one with plane horizontal. Both the coils are connected in series with a dc supply. The coils are found to attract each other with a force. Which one of the following statements is incorrect?

(A) Both the coils carry currents in the same direction.

(B) Coils will attract each other, even if the supply is an ac source.

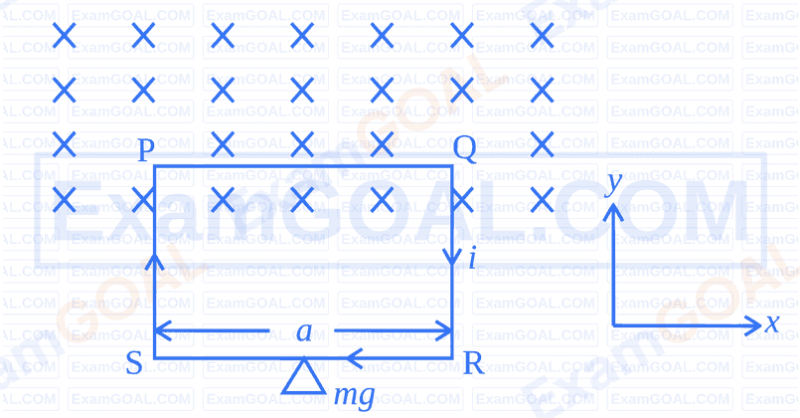
(C) Force is proportional to d^{-1}

(D) Force is proportional to d^{-2}

Q172. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

A rectangular loop of wire, supporting a mass m, hangs with one end in a uniform magnetic field $\vec{B}$ pointing into the plane of the paper. A clockwise current is set up such that $i > mg/Ba$, where a is the width of the loop. Then :



(A) The weight rises due to a vertical force caused by the magnetic field and work is done on the system.

(B) The weight does not rise due to a vertical force caused by the magnetic field and work is done on the system.

(C) The weight rises due to a vertical force caused by the magnetic field but no work is done on the system.

(D) The weight rises due to a vertical force caused by the magnetic field and work is extracted from the magnetic field.

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Q173. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

A particle of charge 16×10^{-16} C moving with velocity 10 ms^{-1} along x-axis enters a region where magnetic field of induction $\vec{B}$ is along the y-axis and an electric field of magnitude 10^4 Vm^{-1} is along the negative z-axis. If the charged particle continues moving along x-axis, the magnitude of $\vec{B}$ is :

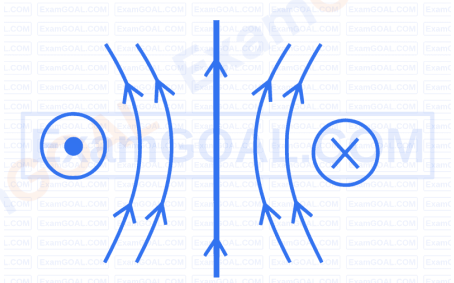
(A) 16×10^3 Wb m^{-2} (B) 2×10^3 Wb m^{-2} (C) 1×10^3 Wb m^{-2} (D) 4×10^3 Wb m^{-2}

Q174. JEE Main 2013 (Online) 22th April Morning Slot

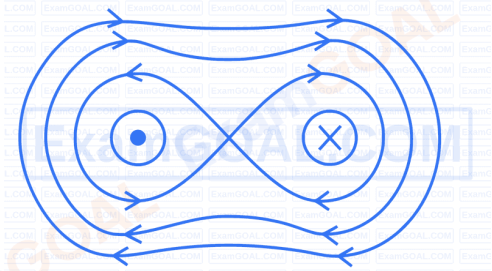
Open solution

Choose the correct sketch of the magnetic field lines of a circular current loop shown by the dot and the cross $\otimes$.

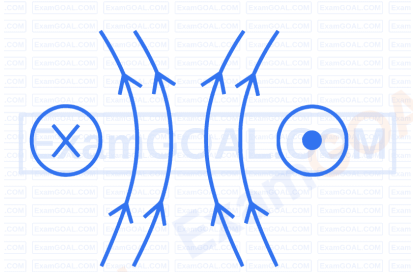
(A)



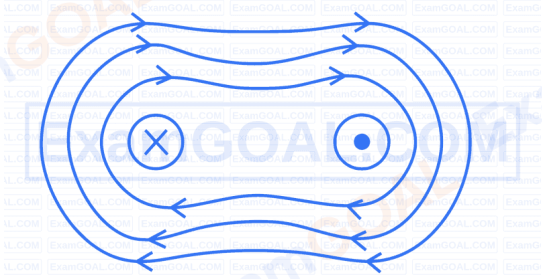
(B)



(C)



(D)



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Q175. JEE Main 2013 (Online) 22th April Morning Slot

A current i is flowing in a straight conductor of length L. The magnetic induction at a point on its axis at a distance $\frac{L}{4}$ from its centre will be :

(A) Zero (B) $\frac{\mu_0 i}{2\pi L}$ (C) $\frac{\mu_0 i}{\sqrt{2}L}$ (D) $\frac{4\mu_0 i}{\sqrt{5}\pi L}$

Open solution

Q176. JEE Main 2013 (Online) 9th April Morning Slot

An electric current is flowing through a circular coil of radius R. The ratio of the magnetic field at the centre of the coil and that at a distance $2\sqrt{2}R$ from the centre of the coil and on its axis is :

(A) $2\sqrt{2}$ (B) 27 (C) 36 (D) 8

Open solution

Q177. AIEEE 2012 (Online) 26th May

Currents of 10 ampere and 2 ampere are passed through two parallel thin wires A and B respectively in opposite directions. Wire A is infinitely long and the length of the wire B is $2\ \mathrm{m}$. The force acting on the conductor B, which is situated at $10\ \mathrm{cm}$ distance from A will be

(A) $8 \times 10^{-5}\ \mathrm{N}$ (B) $5 \times 10^{-5}\ \mathrm{N}$ (C) $8\pi \times 10^{-7}\ \mathrm{N}$ (D) $4\pi \times 10^{-7}\ \mathrm{N}$

Open solution

Q178. AIEEE 2012 (Online) 26th May

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: A charged particle is moving at right angle to a static magnetic field. During the motion the kinetic energy of the charge remains unchanged.
Statement 2: Static magnetic field exert force on a moving charge in the direction perpendicular to the magnetic field.

(A) Statement 1 is false, Statement 2 is true.

(B) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

Open solution

Q179. AIEEE 2012 (Online) 19th May

A proton and a deuteron are both accelerated through the same potential difference and enter in a magnetic field perpendicular to the direction of the field. If the deuteron follows a path of radius R, assuming the neutron and proton masses are nearly equal, the radius of the proton's path will be

(A) $\sqrt{2}R$ (B) $\frac{R}{\sqrt{2}}$ (C) $\frac{R}{2}$ (D) R

Open solution

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Q180. AIEEE 2012 (Online) 12th May

Open solution

The magnetic force acting on charged particle of charge $2\ \mu\text{C}$ in magnetic field of $2\ \text{T}$ acting in y-direction, when the particle velocity is $(2\hat{i} + 3\hat{j}) \times 10^6\ \text{m s}^{-1}$ is

(A) $8\ \text{N}$ in z-direction

(B) $8\ \text{N}$ in y-direction

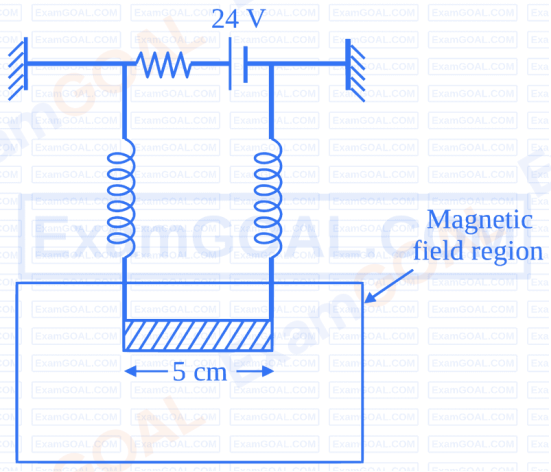
(C) $4\ \text{N}$ in y-direction

(D) $4\ \text{N}$ in z-direction

Q181. AIEEE 2012 (Online) 12th May

Open solution

The circuit in figure consists of wires at the top and bottom and identical springs as the left and right sides. The wire at the bottom has a mass of $10\ \text{g}$ and is $5\ \text{cm}$ long. The wire is hanging as shown in the figure. The springs stretch $0.5\ \text{cm}$ under the weight of the wire and the circuit has a total resistance of $12\ \Omega$. When the lower wire is subjected to a static magnetic field, the springs, stretch an additional $0.3\ \text{cm}$. The magnetic field is



(A) $0.6\ \text{T}$ and directed out of page

(B) $1.2\ \text{T}$ and directed into the plane of page

(C) $0.6\ \text{T}$ and directed into the plane of page

(D) $1.2\ \text{T}$ and directed out of page

Q182. AIEEE 2012 (Online) 7th May

Open solution

The velocity of certain ions that pass undeflected through crossed electric field $E = 7.7\ \text{kV/m}$ and magnetic field $B = 0.14\ \text{T}$ is

(A) $18\ \text{km/s}$

(B) $77\ \text{km/s}$

(C) $55\ \text{km/s}$

(D) $1078\ \text{km/s}$

Q183. AIEEE 2012

Open solution

Proton, deuteron and alpha particle of same kinetic energy are moving in circular trajectories in a constant magnetic field. The radii of proton, deuteron and alpha particle are respectively r_p, r_d and r_α. Which one of the following relations is correct?

(A) $r_\alpha = r_p = r_d$

(B) $r_\alpha = r_p < r_d$

(C) $r_\alpha > r_d > r_p$

(D) $r_\alpha = r_d > r_p$

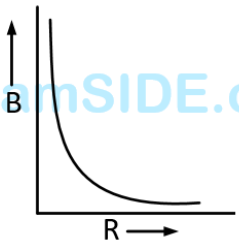
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Q184. AIEEE 2012

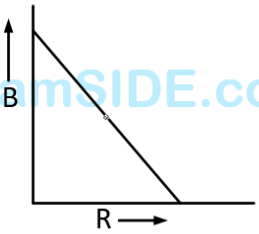
Open solution

A charge Q is uniformly distributed over the surface of non-conducting disc of radius R. The disc rotates about an axis perpendicular to its plane and passing through its center with an angular velocity ω. As a result of this rotation a magnetic field of induction B is obtained at the center of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the center of the disc will be represented by the figure :

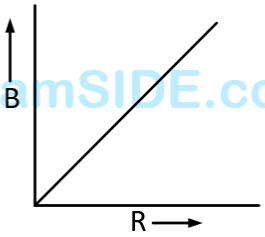
(A)



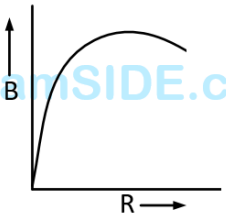
(B)



(C)



(D)



Q185. AIEEE 2011

Open solution

A current I flows in an infinitely long wire with cross section in the form of a semi-circular ring of radius R. The magnitude of the magnetic induction along its axis is:

(A) $\frac{\mu_0 I}{2\pi^2 R}$ (B) $\frac{\mu_0 I}{2\pi R}$ (C) $\frac{\mu_0 I}{4\pi R}$ (D) $\frac{\mu_0 I}{\pi^2 R}$

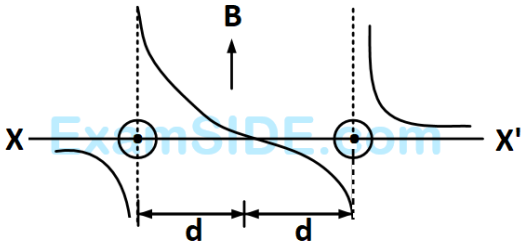
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Q186. AIEEE 2010

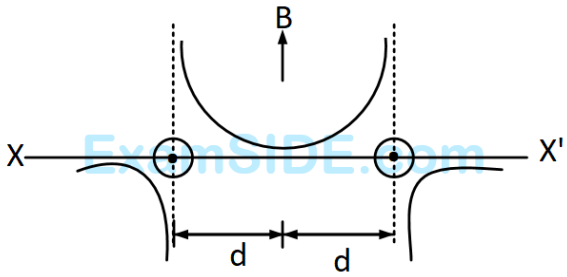
Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by

Open solution

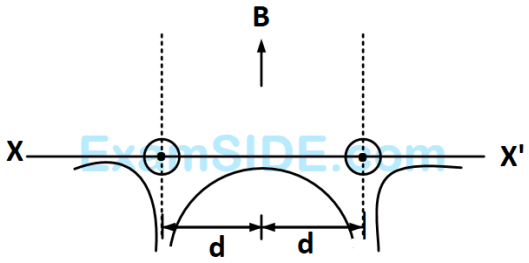
(A)



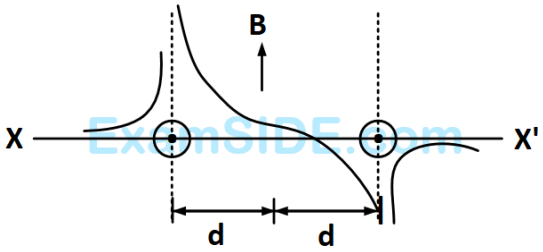
(B)



(C)



(D)

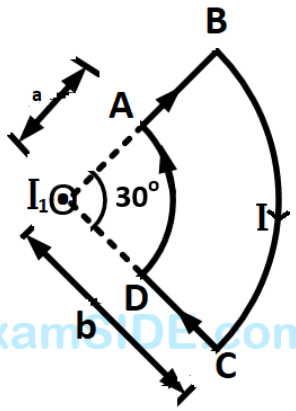


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Q187. AIEEE 2009

Open solution

A current loop $ABCD$ is held fixed on the plane of the paper as shown in the figure. The arcs BC (radius $= b$) and DA (radius $= a$) of the loop are joined by two straight wires AB and CD. A steady current I is flowing in the loop. Angle made by AB and CD at the origin O is 30°. Another straight thin wire steady current I_1 flowing out of the plane of the paper is kept at the origin.



ExamSIDE.com

The magnitude of the magnetic field (B) due to the loop $ABCD$ at the origin (O) is :

(A) $\frac{\mu_0 I(b-a)}{24ab}$

(B) $\frac{\mu_0 I}{4\pi}\left[\frac{b-a}{ab}\right]$

(C) $\frac{\mu_0 I}{4\pi}\left[2(b-a)+\pi/3(a+b)\right]$

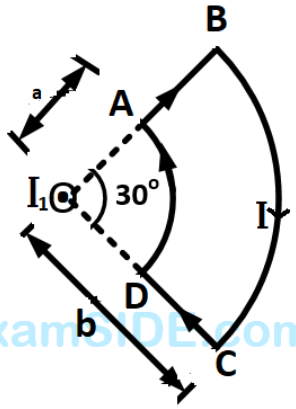
(D) zero

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Q188. AIEEE 2009

A current loop $ABCD$ is held fixed on the plane of the paper as shown in the figure. The arcs BC (radius $= b$) and DA (radius $= a$) of the loop are joined by two straight wires AB and CD. A steady current I is flowing in the loop. Angle made by AB and CD at the origin O is 30°. Another straight thin wire steady current I_1 flowing out of the plane of the paper is kept at the origin.

Open solution



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Due to the presence of the current I_1 at the origin:

(A) The forces on AD are BC are zero.

(B) The magnitude of the net force on the loop is given by $\frac{I_1 I}{4\pi}\mu_0\left[2\left(b-a\right)+\pi/3\left(a+b\right)\right]$.

(C) The magnitude of the net force on the loop is given by $\frac{\mu_0 I I_1}{24ab}\left(b-a\right)$.

(D) The forces on AB and DC are zero.

Q189. AIEEE 2008

A horizontal overhead powerline is at height of $4m$ from the ground and carries a current of $100A$ from east to west. The magnetic field directly below it on the ground is $\left(\mu_0 = 4\pi \times 10^{-7}\ Tm\ A^{-1}\right)$

Open solution

(A) $2.5 \times 10^{-7}\,T$ southward

(B) $5 \times 10^{-6}\,T$ northward

(C) $5 \times 10^{-6}\,T$ southward

(D) $2.5 \times 10^{-7}\,T$ northward

Q190. AIEEE 2007

A long straight wire of radius a carries a steady current i. The current is uniformly distributed across its cross section. The ratio of the magnetic field at $a/2$ and $2a$ is

Open solution

(A) $1/2$ (B) $1/4$ (C) 4 (D) 1

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Q191. AIEEE 2007

A current I flows along the length of an infinitely long, straight, thin walled pipe. Then

(A) the magnetic field at all points inside the pipe is the same, but not zero

(B) the magnetic field is zero only on the axis of the pipe

(C) the magnetic field is different at different points inside the pipe

(D) the magnetic field at any point inside the pipe is zero

Open solution

Q192. AIEEE 2007

A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then

(A) $\vec{v} = \vec{B} \times \vec{E}/E^2$

(B) $\vec{v} = \vec{E} \times \vec{B}/B^2$

(C) $\vec{v} = \vec{B} \times \vec{E}/B^2$

(D) $\vec{v} = \vec{E} \times \vec{B}/E^2$

Open solution

Q193. AIEEE 2007

A charged particle moves through a magnetic field perpendicular to its direction. Then

(A) Kinetic energy changes but the momentum is constant

(B) the momentum changes but the kinetic energy is constant

(C) both momentum and kinetic energy of the particle are not constant

(D) both momentum and kinetic energy of the particle are constant

Open solution

Q194. AIEEE 2007

Two identical conducting wires AOB and COD are placed at right angles to each other. The wire AOB carries an electric current I_1 and COD carries a current I_2. The magnetic field on a point lying at a distance d from O, in a direction perpendicular to the plane of the wires AOB and COD , will be given by

(A) $\frac{\mu_0}{2\pi d}\left(I_1^2 + I_2^2\right)$

(B) $\frac{\mu_0}{2\pi}\left(\frac{I_1+I_2}{d}\right)^{\frac{1}{2}}$

(C) $\frac{\mu_0}{2\pi d}\left(I_1^2 + I_2^2\right)^{\frac{1}{2}}$

(D) $\frac{\mu_0}{2\pi d}\left(I_1 + I_2\right)$

Open solution

Q195. AIEEE 2006

In a region, steady and uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in this region. The path of the particle will be a

(A) helix (B) straight line (C) ellipse (D) circle

Open solution

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Q196. AIEEE 2006

A long solenoid has 200 turns per cm and carries a current i. The magnetic field at its center is $6.28 \times 10^{-2}\ Weber/m^2$. Another long solenoid has 100 turns per cm and it carries a current $\frac{i}{3}$. The value of the magnetic field at its center is

Open solution

(A) $1.05 \times 10^{-2}\ Weber/m^2$
(B) $1.05 \times 10^{-5}\ Weber/m^2$
(C) $1.05 \times 10^{-3}\ Weber/m^2$
(D) $1.05 \times 10^{-4}\ Weber/m^2$

Q197. AIEEE 2005

Two thin, long, parallel wires, separated by a distance $'d'$ carry a current of $'i'\ A$ in the same direction. They will

Open solution

(A) repel each other with a force of $\mu_0 i^2/(2\pi d)$
(B) attract each other with a force of $\mu_0 i^2/(2\pi d)$
(C) repel each other with a force of $\mu_0 i^2/(2\pi d^2)$
(D) attract each other with a force of $\mu_0 i^2/(2\pi d^2)$

Q198. AIEEE 2005

Two concentric coils each of radius equal to $2\ \pi\ cm$ are placed at right angles to each other. 3 ampere and 4 ampere are the currents flowing in each coil respectively . The magnetic induction in Weber / m^2 at the center of the coils will be
$(\mu = 4\pi \times 10^{-7} Wb/A.m)$

Open solution

(A) 10^{-5}
(B) 12×10^{-5}
(C) 7×10^{-5}
(D) 5×10^{-5}

Q199. AIEEE 2005

A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B. The time taken by the particle to complete one revolution is

Open solution

(A) $\frac{2\pi q^2 B}{m}$
(B) $\frac{2\pi mq}{B}$
(C) $\frac{2\pi m}{qB}$
(D) $\frac{2\pi qB}{m}$

Q200. AIEEE 2005

A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected along the direction of the fields with a certain velocity then

Open solution

(A) its velocity will increase
(B) Its velocity will decrease
(C) it will turn towards left of a direction of motion
(D) it will turn towards right of direction of motion

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Q201. AIEEE 2004

A current i ampere flows along an infinitely long straight thin walled tube, then the magnetic induction at any point inside the tube is

(A) $\frac{\mu_0}{4\pi}, \frac{2i}{r}$ tesla
(B) zero
(C) infinite
(D) $\frac{2i}{r}$ tesla

Open solution

Q202. AIEEE 2004

A long wire carries a steady current. It is bent into a circle of one turn and the magnetic field at the centre of the coil is B. It is then bent into a circular loop of n turns. The magnetic field at the center of the coil will be

(A) $2n\ B$
(B) $n^2\ B$
(C) nB
(D) $2n^2\ B$

Open solution

Q203. AIEEE 2004

The magnetic field due to a current carrying circular loop of radius $3\ cm$ at a point on the axis at a distance of $4\ cm$ from the centre is $54\ \mu T$. What will be its value at the center of loop?

(A) $125\ \mu T$
(B) $150\ \mu T$
(C) $250\ \mu T$
(D) $75\ \mu T$

Open solution

Q204. AIEEE 2004

Two long conductors, separated by a distance d carry current I_1 and I_2 in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to $3d$. The new value of the force between them is

(A) $-\frac{2F}{3}$
(B) $\frac{F}{3}$
(C) $-2F$
(D) $-\frac{F}{3}$

Open solution

Q205. AIEEE 2003

A particle of mass M and charge Q moving with velocity $\vec{v}$ describes a circular path of radius R when subjected to a uniform transverse magnetic field of induction B. The net work done by the field when the particle completes one full circle is

(A) $\left(\frac{Mv^2}{R}\right)2\pi R$
(B) zero
(C) $B\ Q\ 2\pi R$
(D) $B\,Qv\,2\pi R$

Open solution

Q206. AIEEE 2003

A particle of charge -16×10^{-18} coulomb moving with velocity $10ms^{-1}$ along the x-axis enters a region where a magnetic field of induction B is along the y-axis, and an electric field of magnitude $10^4 V/m$ is along the negative z-axis. If the charged particle continues moving along the x-axis, the magnitude of B is

(A) $10^3 Wb/m^2$
(B) $10^5 Wb/m^2$
(C) $10^{16} Wb/m^2$
(D) $10^{-3} Wb/m^2$

Open solution

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Q207. AIEEE 2002

If a current is passed through a spring then the spring will

(A) expand (B) compress (C) remain same (D) none of these

Open solution

Q208. AIEEE 2002

If in a circular coil A of radius R, current I is flowing and in another coil B of radius $2R$ a current $2I$ is flowing, then the ratio of the magnetic fields B_A and B_B, produced by them will be

(A) 1 (B) 2 (C) $1/2$ (D) 4

Open solution

Q209. AIEEE 2002

If an electron and a proton having same momentum enter perpendicular to a magnetic field, then

(A) curved path of electron and proton will be same (ignoring the sense of revolution)

(B) they will move undeflected

(C) curved path of electron is more curved than that of the proton

(D) path of proton is more curved.

Open solution

Q210. AIEEE 2002

The time period of a charged particle undergoing a circular motion in a uniform magnetic field is independent of its

(A) speed (B) mass

(C) charge (D) magnetic induction

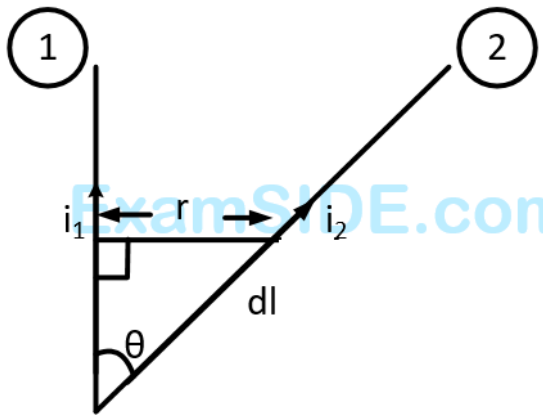
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Q211. AIEEE 2002

Wires 1 and 2 carrying currents i_1 and i_2 respectively are inclined at an angle θ to each other. What is the force on a small element dl of wire 2 at a distance of r from wire 1 (as shown in figure) due to the magnetic field of wire 1?

Open solution



(A) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl\, \tan\theta$

(B) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl\, \sin\theta$

(C) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl\, \cos\theta$

(D) $\frac{\mu_0}{4\pi r} i_1 i_2\, dl\, \sin\theta$

Numerical 51 questions

Q212. JEE Main 2026 (Online) 8th April Evening Shift

A 5 mg particle carrying a charge of $5\pi \times 10^{-6}\text{C}$ is moving with velocity of $(3\hat{i} + 2\hat{k}) \times 10^{-2}\ \text{m/s}$ in a region having magnetic field $\vec{B} = 0.1\hat{k}\ \text{Wb/m}^2$. It moves a distance of α meter along $\hat{k}$ when it completes 5 revolutions. The value of α is ____.

Open solution

Q213. JEE Main 2026 (Online) 6th April Evening Shift

A moving coil of galvanometer when shunted with 2Ω resistance gives a full scale deflection for a current of 500 mA . When a resistance of 470Ω is connected in series it gives a full scale deflection for 10 V potential applied on it. The value of resistance of galvanometer coil is ____ Ω.

Open solution

Q214. JEE Main 2026 (Online) 5th April Morning Shift

The charged particle moving in a uniform magnetic field of $(3\hat{i} + 2\hat{j})\text{T}$ has an acceleration $\left(4\hat{i} - \frac{x}{2}\hat{j}\right)\text{m/s}^2$. The value of x is

Open solution

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Q215. JEE Main 2026 (Online) 4th April Evening Shift

A circular coil of radius 2 cm and 125 turns carries a current of 1 A . The coil is placed in a uniform magnetic field of magnitude 0.4 T . The axis of the coil makes an angle of 30° with the direction of the magnetic field. The torque acting on the coil is $\alpha \times 10^{-4}\ \mathrm{N.m}$. The value of α is ____ .

$(\pi = 3.14)$

Open solution

Q216. JEE Main 2026 (Online) 2nd April Morning Shift

1 μC charge moving with velocity $\vec{v} = (\hat{i} - 2\hat{j} + 3\hat{k})$ m/s in the region of magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 5\hat{k})$ T. The magnitude of force acting on it is $\sqrt{\alpha} \times 10^{-6}$ N. The value of α is _____.

Open solution

Q217. JEE Main 2025 (Online) 4th April Evening Shift

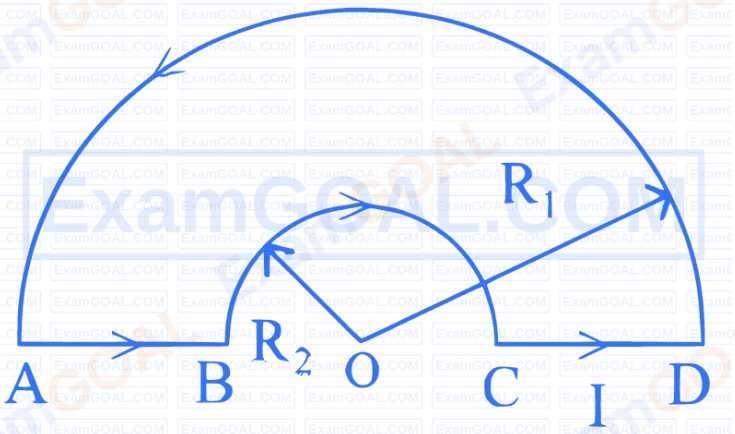
A particle of charge $1.6\mu\mathrm{C}$ and mass $16\mu\ \mathrm{g}$ is present in a strong magnetic field of 6.28 T . The particle is then fired perpendicular to magnetic field. The time required for the particle to return to original location for the first time is _____ s. $(\pi = 3.14)$

Open solution

Q218. JEE Main 2025 (Online) 3rd April Morning Shift

A loop ABCDA , carrying current $\mathrm{I} = 12\ \mathrm{A}$, is placed in a plane, consists of two semi-circular segments of radius $R_1 = 6\pi\ \mathrm{m}$ and $\mathrm{R}_2 = 4\pi\ \mathrm{m}$. The magnitude of the resultant magnetic field at center O is $\mathrm{k} \times 10^{-7}\ \mathrm{T}$. The value of k is_____.

(Given $\mu_0 = 4\pi \times 10^{-7}\mathrm{TmA}^{-1}$)



Open solution

Q219. JEE Main 2025 (Online) 3rd April Morning Shift

A 4.0 cm long straight wire carrying a current of 8 A is placed perpendicular to a uniform magnetic field of strength 0.15 T . The magnetic force on the wire is _____mN .

Open solution

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Q220. JEE Main 2025 (Online) 29th January Evening Shift

The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} Tesla. If the solenoid carries a current of 0.29 A , then the length of the solenoid is _____ πcm.

Open solution

Q221. JEE Main 2025 (Online) 24th January Evening Shift

A tightly wound long solenoid carries a current of 1.5 A . An electron is executing uniform circular motion inside the solenoid with a time period of 75 ns . The number of turns per metre in the solenoid is _____.

[Take mass of electron $m_e = 9 \times 10^{-31}$ kg, charge of electron $|q_e| = 1.6 \times 10^{-19}$C, $\mu_0 = 4\pi \times 10^{-7} \frac{N}{A^2}$, 1 ns $= 10^{-9}$ s]

Open solution

Q222. JEE Main 2025 (Online) 24th January Morning Shift

A current of 5 A exists in a square loop of side $\frac{1}{\sqrt{2}}$ m. Then the magnitude of the magnetic field B at the centre of the square loop will be $p \times 10^{-6}$ T. where, value of p is _____ [Take $\mu_0 = 4\pi \times 10^{-7}$ T mA^{-1}].

Open solution

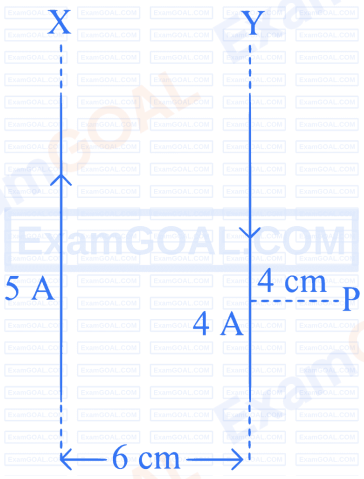
Q223. JEE Main 2025 (Online) 22nd January Evening Shift

A proton is moving undeflected in a region of crossed electric and magnetic fields at a constant speed of 2×10^5 ms^{-1}. When the electric field is switched off, the proton moves along a circular path of radius 2 cm . The magnitude of electric field is $x \times 10^4$ N/C. The value of x is _____. Take the mass of the proton $= 1.6 \times 10^{-27}$ kg.

Open solution

Q224. JEE Main 2025 (Online) 22nd January Evening Shift

Two long parallel wires X and Y, separated by a distance of 6 cm , carry currents of 5 A and 4A, respectively, in opposite directions as shown in the figure. Magnitude of the resultant magnetic field at point P at a distance of 4 cm from wire Y is $x \times 10^{-5}$ T. The value of x is _____ . Take permeability of free space as $\mu_0 = 4\pi \times 10^{-7}$ SI units.



Open solution

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Q225. JEE Main 2024 (Online) 9th April Evening Shift

A straight magnetic strip has a magnetic moment of $44\ \text{Am}^2$. If the strip is bent in a semicircular shape, its magnetic moment will be _____ Am^2.

(given $\pi = \frac{22}{7}$)

Open solution

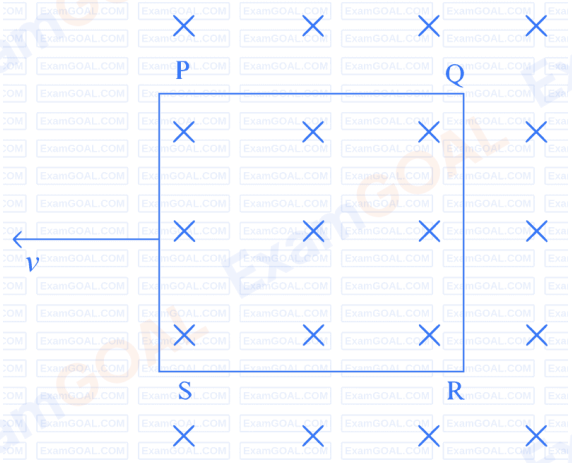
Q226. JEE Main 2024 (Online) 9th April Morning Shift

A square loop of edge length $2\ \text{m}$ carrying current of $2\ \text{A}$ is placed with its edges parallel to the x-y axis. A magnetic field is passing through the x-y plane and expressed as $\vec{B} = B_0(1 + 4x)\hat{k}$, where $B_0 = 5\ \text{T}$. The net magnetic force experienced by the loop is _____ N.

Open solution

Q227. JEE Main 2024 (Online) 8th April Morning Shift

A square loop PQRS having 10 turns, area $3.6 \times 10^{-3}\ \text{m}^2$ and resistance 100Ω is slowly and uniformly being pulled out of a uniform magnetic field of magnitude $\text{B} = 0.5\ \text{T}$ as shown. Work done in pulling the loop out of the field in $1.0\ \text{s}$ is _____ $\times 10^{-6}$ J.



Open solution

Q228. JEE Main 2024 (Online) 8th April Morning Shift

An electron with kinetic energy $5\ \text{eV}$ enters a region of uniform magnetic field of $3\ \mu\text{T}$ perpendicular to its direction. An electric field E is applied perpendicular to the direction of velocity and magnetic field. The value of E, so that electron moves along the same path, is _____ NC^{-1}.

(Given, mass of electron $= 9 \times 10^{-31}\ \text{kg}$, electric charge $= 1.6 \times 10^{-19}\text{C}$)

Open solution

Q229. JEE Main 2024 (Online) 6th April Evening Shift

A coil having 100 turns, area of $5 \times 10^{-3}\ \text{m}^2$, carrying current of $1\ \text{mA}$ is placed in uniform magnetic field of $0.20\ \text{T}$ such a way that plane of coil is perpendicular to the magnetic field. The work done in turning the coil through 90° is _____ μJ.

Open solution

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Q230. JEE Main 2024 (Online) 6th April Morning Shift

A circular coil having 200 turns, $2.5 \times 10^{-4}\ \mathrm{m}^2$ area and carrying $100\mu\mathrm{A}$ current is placed in a uniform magnetic field of $1\ \mathrm{T}$. Initially the magnetic dipole moment $(\vec{M})$ was directed along $\vec{B}$. Amount of work, required to rotate the coil through 90° from its initial orientation such that $\vec{M}$ becomes perpendicular to $\vec{B}$, is _____ μJ.

Open solution

Q231. JEE Main 2024 (Online) 5th April Evening Shift

A solenoid of length $0.5\ \mathrm{m}$ has a radius of $1\ \mathrm{cm}$ and is made up of 'm' number of turns. It carries a current of $5\ \mathrm{A}$. If the magnitude of the magnetic field inside the solenoid is $6.28 \times 10^{-3}\ \mathrm{T}$ then the value of m is _____.

Open solution

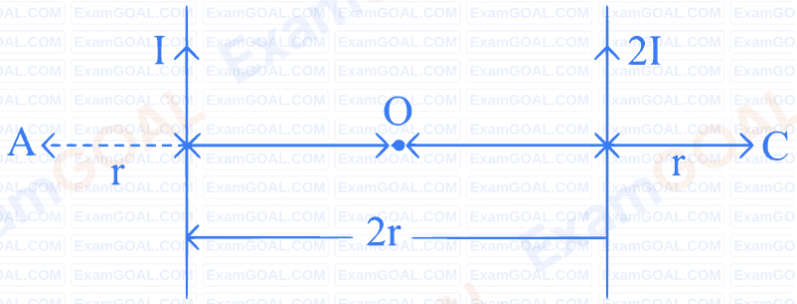
Q232. JEE Main 2024 (Online) 5th April Morning Shift

A 2A current carrying straight metal wire of resistance 1Ω, resistivity $2 \times 10^{-6}\Omega\mathrm{m}$, area of cross-section $10\ \mathrm{mm}^2$ and mass $500\ \mathrm{g}$ is suspended horizontally in mid air by applying a uniform magnetic field $\vec{B}$. The magnitude of B is _____ $\times 10^{-1}\ \mathrm{T}$ (given, $\mathrm{g} = 10\ \mathrm{m/s}^2$).

Open solution

Q233. JEE Main 2024 (Online) 4th April Evening Shift

Two parallel long current carrying wires separated by a distance $2r$ are shown in the figure. The ratio of magnetic field at A to the magnetic field produced at C is $\frac{x}{7}$. The value of x is _____.



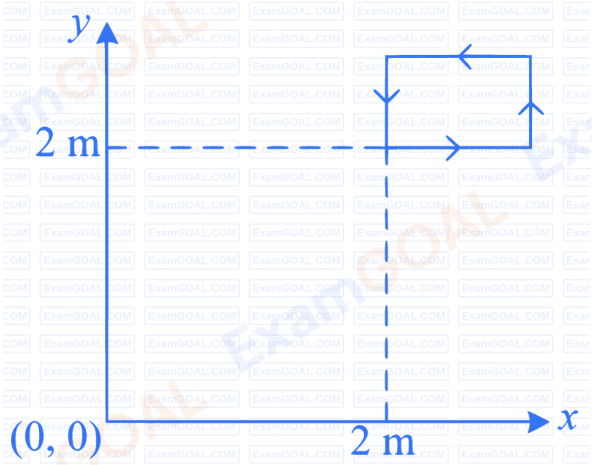
Open solution

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Q234. JEE Main 2024 (Online) 4th April Morning Shift

The magnetic field existing in a region is given by $\vec{B} = 0.2(1 + 2x)\hat{k}$. A square loop of edge $50\ \text{cm}$ carrying 0.5 A current is placed in x-y plane with its edges parallel to the x-y axes, as shown in figure. The magnitude of the net magnetic force experienced by the loop is _____ mN.

Open solution



Q235. JEE Main 2024 (Online) 1st February Evening Shift

A moving coil galvanometer has 100 turns and each turn has an area of $2.0\ \text{cm}^2$. The magnetic field produced by the magnet is $0.01\ \text{T}$ and the deflection in the coil is 0.05 radian when a current of $10\ \text{mA}$ is passed through it. The torsional constant of the suspension wire is $x \times 10^{-5}\ \text{N} - \text{m/rad}$. The value of x is _____ .

Open solution

Q236. JEE Main 2024 (Online) 1st February Morning Shift

A regular polygon of 6 sides is formed by bending a wire of length 4π meter.

If an electric current of $4\pi\sqrt{3}$ A is flowing through the sides of the polygon, the magnetic field at the centre of the polygon would be $x \times 10^{-7}\ \text{T}$.

The value of x is _____.

Open solution

Q237. JEE Main 2024 (Online) 31st January Evening Shift

Two circular coils P and Q of 100 turns each have same radius of $\pi\ \text{cm}$. The currents in P and Q are $1A$ and $2A$ respectively. P and Q are placed with their planes mutually perpendicular with their centers coincide. The resultant magnetic field induction at the center of the coils is $\sqrt{x}\ mT$, where $x =$ _____.

[Use $\mu_0 = 4\pi \times 10^{-7}\ \text{TmA}^{-1}$]

Open solution

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Q238. JEE Main 2024 (Online) 31st January Morning Shift

An electron moves through a uniform magnetic field $\vec{B} = B_0\hat{i} + 2B_0\hat{j}T$. At a particular instant of time, the velocity of electron is $\vec{u} = 3\hat{i} + 5\hat{j}\,\mathrm{m/s}$. If the magnetic force acting on electron is $\vec{F} = 5e\hat{k}N$, where e is the charge of electron, then the value of B_0 is _____ T.

Open solution

Q239. JEE Main 2024 (Online) 30th January Evening Shift

A current of $5\ \mathrm{A}$ flows in a square loop of side $1\ \mathrm{m}$ placed in air. The magnetic field at the centre of the loop is $X\sqrt{2} \times 10^{-7}T$. The value of X is _____.

Open solution

Q240. JEE Main 2024 (Online) 30th January Morning Shift

The horizontal component of earth's magnetic field at a place is $3.5 \times 10^{-5}\ \mathrm{T}$. A very long straight conductor carrying current of $\sqrt{2}\ \mathrm{A}$ in the direction from South east to North West is placed. The force per unit length experienced by the conductor is _____ $\times 10^{-6}\ \mathrm{N/m}$.

Open solution

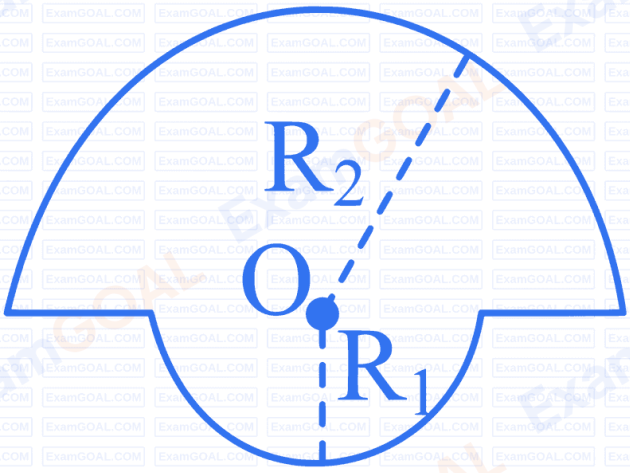
Q241. JEE Main 2024 (Online) 29th January Evening Shift

A charge of $4.0\mu\mathrm{C}$ is moving with a velocity of $4.0 \times 10^{6}\ \mathrm{ms}^{-1}$ along the positive y axis under a magnetic field $\vec{B}$ of strength $(2\hat{k})\mathrm{T}$. The force acting on the charge is $x\hat{i}N$. The value of x is _____.

Open solution

Q242. JEE Main 2024 (Online) 27th January Evening Shift

The magnetic field at the centre of a wire loop formed by two semicircular wires of radii $R_1 = 2\pi\mathrm{m}$ and $R_2 = 4\pi\mathrm{m}$, carrying current $\mathrm{I} = 4\ \mathrm{A}$ as per figure given below is $\alpha \times 10^{-7}\ \mathrm{T}$. The value of α is _____. (Centre O is common for all segments)



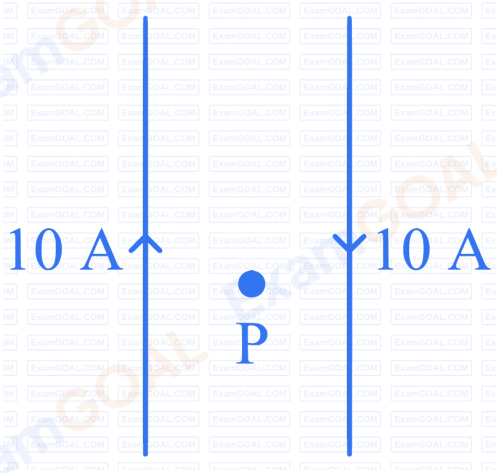
Open solution

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Q243. JEE Main 2024 (Online) 27th January Morning Shift

Two long, straight wires carry equal currents in opposite directions as shown in figure. The separation between the wires is $5.0\ \text{cm}$. The magnitude of the magnetic field at a point P midway between the wires is _____ μT

(Given : $\mu_0 = 4\pi \times 10^{-7}\text{TmA}^{-1}$)



Open solution

Q244. JEE Main 2023 (Online) 15th April Morning Shift

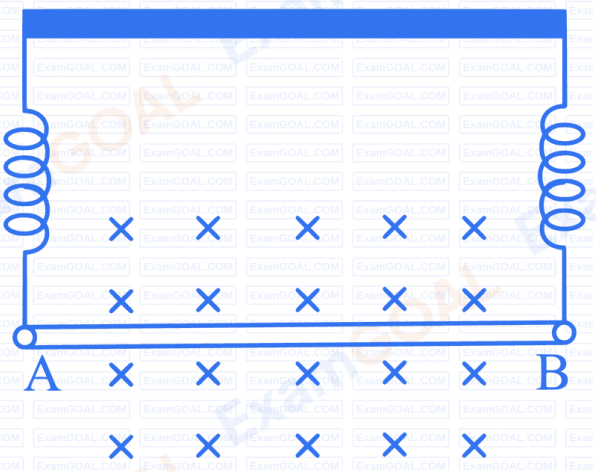
An electron in a hydrogen atom revolves around its nucleus with a speed of $6.76 \times 10^6\ \text{ms}^{-1}$ in an orbit of radius $0.52\ \text{A}^\circ$. The magnetic field produced at the nucleus of the hydrogen atom is _____ T.

Open solution

Q245. JEE Main 2023 (Online) 13th April Evening Shift

A straight wire AB of mass $40\ \text{g}$ and length $50\ \text{cm}$ is suspended by a pair of flexible leads in uniform magnetic field of magnitude $0.40\ \text{T}$ as shown in the figure. The magnitude of the current required in the wire to remove the tension in the supporting leads is _____ A.

(Take $g = 10\ \text{ms}^{-2}$).

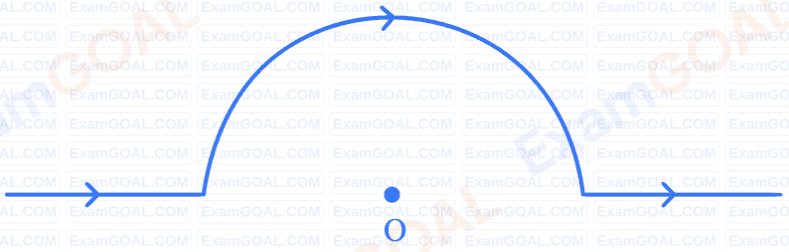


Open solution

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Q246. JEE Main 2023 (Online) 10th April Evening Shift

A straight wire carrying a current of $14\ \mathrm{A}$ is bent into a semi-circular arc of radius $2.2\ \mathrm{cm}$ as shown in the figure. The magnetic field produced by the current at the centre (O) of the arc. is _____ $\times 10^{-4}\ \mathrm{T}$



Open solution

Q247. JEE Main 2023 (Online) 8th April Evening Shift

The ratio of magnetic field at the centre of a current carrying coil of radius r to the magnetic field at distance r from the centre of coil on its axis is $\sqrt{x} : 1$. The value of x is _____

Open solution

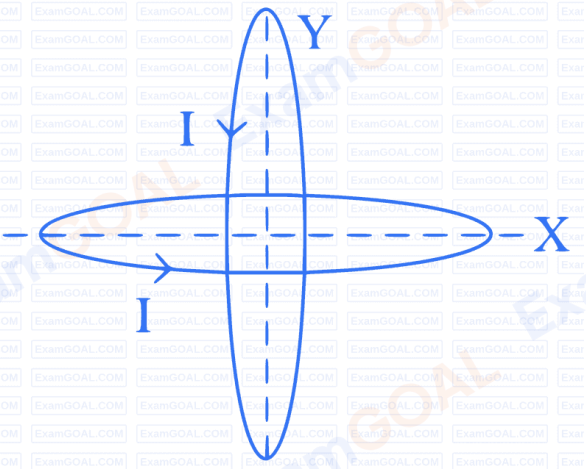
Q248. JEE Main 2023 (Online) 6th April Evening Shift

A proton with a kinetic energy of $2.0\ \mathrm{eV}$ moves into a region of uniform magnetic field of magnitude $\frac{\pi}{2} \times 10^{-3}\ \mathrm{T}$. The angle between the direction of magnetic field and velocity of proton is 60°. The pitch of the helical path taken by the proton is _____ cm. (Take, mass of proton $= 1.6 \times 10^{-27}\ \mathrm{kg}$ and Charge on proton $= 1.6 \times 10^{-19}\mathrm{C}$).

Open solution

Q249. JEE Main 2023 (Online) 6th April Morning Shift

Two identical circular wires of radius $20\ \mathrm{cm}$ and carrying current $\sqrt{2}\ \mathrm{A}$ are placed in perpendicular planes as shown in figure. The net magnetic field at the centre of the circular wires is _____ $\times 10^{-8}\ \mathrm{T}$.



(Take $\pi = 3.14$)

Open solution

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Q250. JEE Main 2023 (Online) 1st February Morning Shift

A charge particle of $2\ \mu C$ accelerated by a potential difference of $100\ V$ enters a region of uniform magnetic field of magnitude $4\ mT$ at right angle to the direction of field. The charge particle completes semicircle of radius $3\ cm$ inside magnetic field. The mass of the charge particle is _____ $\times 10^{-18}$ kg

Open solution

Q251. JEE Main 2023 (Online) 25th January Evening Shift

Two long parallel wires carrying currents 8A and 15A in opposite directions are placed at a distance of 7 cm from each other. A point P is equidistant from both the wires such that the lines joining the point P to the wires are perpendicular to each other. The magnitude of magnetic field at P is _____ $\times 10^{-6}$ T.

(Given : $\sqrt{2} = 1.4$)

Open solution

Q252. JEE Main 2023 (Online) 24th January Evening Shift

A single turn current loop in the shape of a right angle triangle with sides 5 cm, 12 cm, 13 cm is carrying a current of 2 A. The loop is in a uniform magnetic field of magnitude 0.75 T whose direction is parallel to the current in the 13 cm side of the loop. The magnitude of the magnetic force on the 5 cm side will be $\frac{x}{130}$ N. The value of x is _____.

Open solution

Q253. JEE Main 2022 (Online) 29th July Morning Shift

A closely wound circular coil of radius 5 cm produces a magnetic field of $37.68 \times 10^{-4}\ T$ at its center. The current through the coil is _____A.

[Given, number of turns in the coil is 100 and $\pi = 3.14$]

Open solution

Q254. JEE Main 2022 (Online) 29th July Morning Shift

A wire of length $314\ cm$ carrying current of $14\ A$ is bent to form a circle. The magnetic moment of the coil is _____ A $-m^2$. [Given $\pi = 3.14$]

Open solution

Q255. JEE Main 2022 (Online) 28th June Morning Shift

A singly ionized magnesium atom (A = 24) ion is accelerated to kinetic energy 5 keV, and is projected perpendicularly into a magnetic field B of the magnitude 0.5 T. The radius of path formed will be _____ cm.

Open solution

Q256. JEE Main 2022 (Online) 27th June Evening Shift

A deuteron and a proton moving with equal kinetic energy enter into a uniform magnetic field at right angle to the field. If r_d and r_p are the radii of their circular paths respectively, then the ratio $\frac{r_d}{r_p}$ will be $\sqrt{x}$: 1 where x is _____.

Open solution

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Q257. JEE Main 2022 (Online) 26th June Evening Shift

Two 10 cm long, straight wires, each carrying a current of 5A are kept parallel to each other. If each wire experienced a force of 10^{-5} N, then separation between the wires is _____ cm.

Open solution

Q258. JEE Main 2021 (Online) 31st August Evening Shift

A long solenoid with 1000 turns/m has a core material with relative permeability 500 and volume 10^3 cm^3. If the core material is replaced by another material having relative permeability of 750 with same volume maintaining same current of 0.75 A in the solenoid, the fractional change in the magnetic moment of the core would be approximately $\left(\frac{x}{499}\right)$. Find the value of x.

Open solution

Q259. JEE Main 2021 (Online) 27th August Morning Shift

A uniform conducting wire of length 24a and resistance R is wound up as a current carrying coil in the shape of an equilateral triangle of side 'a' and then in the form of a square of side 'a'. The coil is connected to a voltage source V_0. The ratio of magnetic moment of the coils in case of equilateral triangle to that for square is 1 : $\sqrt{y}$ where y is ..... .

Open solution

Q260. JEE Main 2021 (Online) 26th August Evening Shift

If the maximum value of accelerating potential provided by a radio frequency oscillator is 12 kV. The number of revolutions made by a proton in a cyclotron to achieve one sixth of the speed of light is .....

[$m_p = 1.67 \times 10^{-27}$ kg, $e = 1.6 \times 10^{-19}$ C, Speed of light = 3×10^8 m/s]

Open solution

Q261. JEE Main 2021 (Online) 26th August Evening Shift

A coil in the shape of an equilateral triangle of side 10 cm lies in a vertical plane between the pole pieces of permanent magnet producing a horizontal magnetic field 20 mT. The torque acting on the coil when a current of 0.2 A is passed through it and its plane becomes parallel to the magnetic field will be $\sqrt{x} \times 10^{-5}$ Nm. The value of x is .....

Open solution

Q262. JEE Main 2020 (Online) 3rd September Evening Slot

A galvanometer coil has 500 turns and each turn has an average area of 3×10^{-4} m^2 . If a torque of 1.5 Nm is required to keep this coil parallel to a magnetic field when a current of 0.5 A is flowing through it, the strength of the field (in T) is _____.

Open solution

Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q263. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

A cyclotron is working at a frequency of 10 MHz. If the radius of its dees is 60 cm. The maximum kinetic energy of accelerated proton will be :

(Take : e = 1.6×10^{-19} C, $m_p = 1.67 \times 10^{-27}$ kg)

Open solution

(A) 7.4 MeV (B) 14.86 MeV (C) 7.4 GeV (D) 704 GeV

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **B**	3. **A**	4. **A**	5. **B**
6. **A**	7. **A**	8. **D**	9. **B**	10. **B**
11. **B**	12. **A**	13. **B**	14. **D**	15. **B**
16. **A**	17. **A**	18. **C**	19. **A**	20. **A**
21. **C**	22. **B**	23. **B**	24. **B**	25. **B**
26. **B**	27. **A**	28. **C**	29. **D**	30. **C**
31. **C**	32. **C**	33. **B**	34. **D**	35. **A**
36. **B**	37. **B**	38. **D**	39. **C**	40. **C**
41. **A**	42. **B**	43. **A**	44. **C**	45. **B**
46. **A**	47. **A**	48. **A**	49. **C**	50. **A**
51. **C**	52. **D**	53. **A**	54. **A**	55. **D**
56. **D**	57. **C**	58. **C**	59. **D**	60. **C**
61. **B**	62. **C**	63. **A**	64. **C**	65. **C**
66. **D**	67. **A**	68. **B**	69. **B**	70. **A**
71. **B**	72. **B**	73. **B**	74. **B**	75. **D**
76. **B**	77. **C**	78. **C**	79. **C**	80. **D**
81. **C**	82. **C**	83. **A**	84. **B**	85. **D**
86. **A**	87. **C**	88. **A**	89. **D**	90. **A**
91. **A**	92. **B**	93. **D**	94. **C**	95. **B**
96. **C**	97. **B**	98. **D**	99. **A**	100. **C**
101. **A**	102. **A**	103. **A**	104. **D**	105. **B**
106. **A**	107. **C**	108. **D**	109. **A**	110. **A**
111. **C**	112. **D**	113. **C**	114. **C**	115. **D**
116. **C**	117. **D**	118. **C**	119. **D**	120. **D**
121. **B**	122. **D**	123. **A**	124. **C**	125. **B**
126. **A**	127. **D**	128. **D**	129. **B**	130. **B**
131. **D**	132. **D**	133. **A**	134. **D**	135. **C**
136. **B**	137. **C**	138. **C**	139. **C**	140. **A**
141. **D**	142. **A**	143. **B**	144. **B**	145. **D**
146. **D**	147. **D**	148. **D**	149. **D**	150. **C**

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151. **C**	152. **D**	153. **B**	154. **B**	155. **C**
156. **D**	157. **C**	158. **A**	159. **B**	160. **B**
161. **D**	162. **A**	163. **C**	164. **B**	165. **D**
166. **A**	167. **C**	168. **B**	169. **A**	170. **B**
171. **D**	172. **A**	173. **C**	174. **A**	175. **A**
176. **B**	177. **A**	178. **D**	179. **B**	180. **A**
181. **A**	182. **C**	183. **B**	184. **A**	185. **D**
186. **A**	187. **A**	188. **A**	189. **C**	190. **D**
191. **D**	192. **B**	193. **B**	194. **C**	195. **B**
196. **A**	197. **B**	198. **D**	199. **C**	200. **B**
201. **B**	202. **B**	203. **C**	204. **A**	205. **B**
206. **A**	207. **B**	208. **A**	209. **A**	210. **A**
211. **C**				

Numerical

212. **2**	213. **50**	214. **12**	215. **314**	216. **171**
217. **0**	218. **1**	219. **48**	220. **8**	221. **250**
222. **8**	223. **2**	224. **1**	225. **28**	226. **160**
227. **3**	228. **4**	229. **100**	230. **5**	231. **500**
232. **5**	233. **5**	234. **50**	235. **4**	236. **72**
237. **20**	238. **5**	239. **40**	240. **35**	241. **32**
242. **3**	243. **160**	244. **40**	245. **2**	246. **2**
247. **8**	248. **40**	249. **628**	250. **144**	251. **68**
252. **9**	253. **3**	254. **11**	255. **10**	256. **2**
257. **5**	258. **250**	259. **3**	260. **543**	261. **3**
262. **20**				

Out of syllabus

263. **A**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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